Course Structure

R22

Electrical and Electronics Engineering

For
B.Tech 4-Year Degree Course

(Applicable for the students admitted into E1 from the Academic Year 2022-23)

(I – IV Years Syllabus)

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES
Basar, Nirmal, Telangana – 504107

Electrical and Electronics Engineering Dept

S.No	Course Category	Course Type	Standard Credits as per AICTE	Department Credits
1	BSC	Basic Science Course	25	21
2	ESC	Engineering Science Course	24	33
3	HSMC	Humanities and Social Sciences including Management Course	12	10
4	PCC	Program Core Course	48	56
5	PEC	Program Elective Course	18	15
6	OEC	Open Elective Course	18	12
7	MC	Mandatory Course	0	0
8	SIP	Seminar + Internship + Project Work + Comprehensive Viva	15	13
Total			**160**	**160**

Electrical and Electronics Engineering Dept

Course Structure

FIRST YEAR (E1) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1101	Linear algebra and Calculus	BSC	3	1	0	4	4
2	PH1101	Engineering Physics	BSC	3	0	0	3	3
3	PH1701	Engineering Physics Lab	BSC	0	0	3	3	1.5
4	HS1101	English	HSMC	2	0	0	2	2
5	HS1701	English Language Lab	HSMC	0	0	2	2	1
6	BS1101	Environmental science	MC	2	0	0	2	0
7	CE1701	Engineering Graphics	ESC	0	1	4	5	3
8	EE1101	Network Theory-I	PCC	3	0	0	3	3
Total				13	2	9	24	17.5
Induction Programme (Non-Credit)								

FIRST YEAR (E1) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE1201	Network Theory-II	PCC	3	0	0	3	3
2	EE1801	Network Theory Lab	ESC	0	0	3	3	1.5
3	MA1202	Differential equations andLaplace Transforms	BSC	3	1	0	4	4
4	CY1201	Engineering Chemistry	BSC	3	0	0	3	3
5	CY1801	Engineering Chemistry Lab	BSC	0	0	3	3	1.5
6	CS1202	Programming for ProblemSolving	ESC	3	0	0	3	3
7	CS1802	Programming for ProblemSolving Lab	ESC	0	0	3	3	1.5
8	ME1802	Engineering Workshop	ESC	0	1	2	3	2
9	BM1205	Constitution of India	MC	2	0	0	2	0
10	EC1203	Analog Electronic Circuits-I	ESC	3	0	0	3	3
Total				17	2	11	30	22.5

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SECOND YEAR (E2) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE2101	Electrical Machines-I	PCC	3	1	0	4	4
2	EC2104	Analog Electronic Circuits-II	ESC	3	0	0	3	3
3	EC2103	Electromagnetic Fields	PCC	3	1	0	4	4
4	EC2701	Analog Electronic Circuits Lab	ESC	0	0	2	2	1
5	ME2105	Engineering Mechanics	ESC	3	1	0	4	4
6	MA2105	Vector Calculus andComplex analysis	BSC	3	1	0	4	4
7	EE2701	Electrical Machines-I Lab	PCC	0	0	3	3	1.5
8	CS2301	Data Structures **(Open Elective-I)**	OEC	3	0	0	3	3
Total				15	4	5	24	24.5

SECOND YEAR (E2) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE2201	Electrical Machines-II	PCC	3	1	0	4	4
2	EE2202	Power Electronics	PCC	3	1	0	4	4
3	EC2205	Digital Electronics	ESC	3	0	0	3	3
4	EE2203	Power Systems-I	PCC	3	0	0	3	3
5	EC2206	Signals and Systems	ESC	2	1	0	3	3
6	EE2801	Electrical Machines-II Lab	PCC	0	0	3	3	1.5
7	EE2802	Power Electronics Lab	PCC	0	0	3	3	1.5
8	EC2803	Digital Electronics Lab	ESC	0	0	2	2	1
9	HS2201	Essence of Indian Traditional Knowledge	MC	2	0	0	2	0
Total				14	3	10	27	21

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THIRD YEAR (E3) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE3101	Power Systems-II	PCC	3	1	0	4	4
2	EE3102	Control Systems	PCC	3	1	0	4	4
3	EE3103	Electrical Measurementsand Instrumentation	PCC	3	1	0	4	4
4	EC3104	Micro Processors & Micro Controllers	ESC	3	0	0	3	3
5	EE3701	Control Systems Lab	PCC	0	0	2	2	1
6	EE3702	Electrical Measurementsand Instrumentation Lab	PCC	0	0	2	2	1
7	EC3703	Micro Processors & Micro Controllers Lab	ESC	0	0	2	2	1
8	HS3101	Corporate Communication	HSMC	0	0	2	2	1
Total				12	3	6	21	19

THIRD YEAR (E3) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE3201	Power Semiconductor Drives	PCC	4	0	0	4	4
2	EE3202	Power Systems Operationand Control	PCC	3	1	0	4	4
3	EE3801	Power Systems Lab	PCC	0	0	2	2	1
4	EE3803	Electrical Simulation Lab	PCC	0	0	3	3	1.5
5	EE321X	Program Elective-I	PEC	3	0	0	3	3
6	EE322X	Program Elective-II	PEC	3	0	0	3	3
7	EE323X	Program Elective-III	PEC	3	0	0	3	3
8	EE3902	Mini Project	SIP	0	0	2	2	1
9	CS3401	Open Elective-II(Oops)	OEC	3	0	0	0	3
		Total		19	1	9	26	23.5

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FOURTH YEAR (E4) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE4101	Utilization of Electrical Energy	PCC	3	0	0	3	3
2	EE4701	Energy Systems Lab	PCC	0	0	2	2	1
3	EE4142	Program Elective-IV (ANN)	PEC	3	0	0	3	3
4	EE4154	Program Elective-V (HEV)	PEC	3	0	0	3	3
5	EE4901	Major Project-I	SIP	0	0	8	8	4
6	EE4900	Summer Internship	SIP	0	0	0	0	1
7	HS3201	Communication Skills	HSMC	1	0	0	1	1
8	BM4107	MEFA	HSMC	3	0	0	3	3
12	EE4000	Comprehensive Viva	SIP	0	0	0	0	1
Total				13	0	10	23	20

FOURTH YEAR (E4) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	XXXXX	Open Elective-III	OEC	3	0	0	3	3
2	XXXXX	Open Elective-IV	OEC	3	0	0	3	3
3	EE4902	Major Project-II	SIP	0	0	12	12	6
Total				6	0	12	12	12

Total Credits - 160

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FIRST YEAR (E1) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1101	Linear algebra and Calculus	BSC	3	1	0	4	4
2	PH1101	Engineering Physics	BSC	3	0	0	3	3
3	PH1701	Engineering Physics Lab	BSC	0	0	3	3	1.5
4	HS1101	English	HSMC	2	0	0	2	2
5	HS1701	English Language Lab	HSMC	0	0	2	2	1
6	BS1101	Environmental science	MC	2	0	0	2	0
7	CE1701	Engineering Graphics	ESC	0	1	4	5	3
8	EE1101	Network Theory-I	PCC	3	0	0	3	3
Total				13	2	9	24	17.5
Induction Programme (Non-Credit)								

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MA1101 **Linear Algebra and Calculus**

L	T	P	C
3	1	0	4

Internals: 40 Marks
Externals: 60 marks

Course Objectives: To learn

- Types of matrices and their properties.
- Concept of a rank of the matrix and applying this concept to know the consistency and solving the system of linear equations.
- Concept of Eigen values and Eigen vectors and to reduce the quadratic form to canonical form.
- Concept of Sequence.
- Concept of nature of series.
- Geometrical approach to the mean values theorems and their application to the mathematical problems.
- Definition of improper integrals using Beta and Gamma functions.
- Partial differentiation, concept of total derivative.
- Finding maxima and minima of function of two and three variables.

Course outcomes: After learning the concepts of this paper the student must be able to
Write the matrix representation of set of linear equations and to analyze the solution of the system of equations.

- Find the Eigen values and Eigen vectors.
- Reduce the quadratic form to canonical form using orthogonal transformations.
- Analyze the nature of sequence and series.
- Solve the applications on the mean value theorems.
- Improper integrals using Beta and Gamma functions.

UNIT-I: Matrix Theory (10)

Types of Matrices, Symmetric, Hermitian, Skew-Symmetry, Skew-Hermitian, Orthogonal matrices, Unitary matrices; Elementary row and column operations on a matrix, Rank of a matrix by Echelon form and Normal form, Inverse of a Non-singular matrix by Gauss-Jordan method; Consistency and solutions of system of linear equations using elementary operations.

UNIT-II: Eigen values and Eigen vectors (10)

Linear Transformation and Orthogonal Transformation; Characteristic roots and vectors of a matrix; Diagonalization of a matrix; Cayley-Hamilton theorem(without proof) ; finding inverse

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and power of a matrix by Caylay-Hamilton Theorem; Quadratic forms and Nature of the Quadratic forms.

UNIT-III: Sequences & Series (10)

Definition of a sequence, limit; Convergent, Divergent and Oscillatory sequences. Convergent, Divergent and Oscillatory Series; Series of positive terms; Comparison test, p-test, D-Alembert's ratio test; Raabe's test; Cauchy's Integral test; Cauchy's root test; Logarithmic test. Alternating series; Leibnitz test; Alternating Convergent series; Absolute and conditionally convergence.

UNIT-IV: Calculus (10)

Mean value theorems: Roll's theorem, Lagrange's Mean value theorem, Cauchy's Mean value theorem; Taylor's and Maclaurin's series with remainders, Expansions; Definition of Improper Integrals and Beta and Gamma functions and their applications.

UNIT-V: Multivariable Calculus (10)

(Partial Differentiation and applications): Functions of two variables, Definitions of Limits and continuity, Partial Differentiation; Euler's theorem; Total Derivative; Jacobian; Functional dependence and independence; Maxima and minima of functions of several variables (two and three variables) using Lagrange Multipliers.

TEXTBOOKS:

1. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8th Edition,
2. R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House.2008
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.

REFERENCES:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson,Reprint,2002.
2. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11thReprint, 2010.
3. N.P. bail and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint,2008.

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PH1101 **ENGINEERING PHYSICS**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Unit I: Vectors and Mathematical Physics (14)

Gradient, Divergence, Curl and its applications, Line, surface and volume integrals, Stokes and Gauss theorem: Applications, Curvilinear Coordinates: Polar, Cylindrical and spherical co-ordinates, Problems. Electrodynamics before Maxwell, Fixing of Ampere's Law, Maxwell Equations in Matter, Boundary Conditions, Continuity Equation, Poynting Theorem, Wave equation for E and B, Monochromatic Plane Waves, Energy and Momentum in EM Waves. Propagation in Linear Media, Reflection and Transmission at Normal Incidence. EM Waves in Conductors, Reflection at Conducting Surface.

Unit II: Quantum Mechanics (8)

Introduction to Quantum Mechanics, De-Broglie's waves and uncertainty principle, Wave Function and its Significance, Time dependant and time independent Schrodinger wave equations, Particle in a box - Problems.

Unit III: Electron Structure of Solids (10)

Introduction to Crystallography, Bravais Lattices and crystal systems, Atomic Packing, Atomic Radii, Crystal Structures (SC, BCC and FCC), Miller Indices, Classical Free electron Theory, Kronig Penny model (E vs K), Band theory of solids.

Unit IV: Semiconductor Physics (8)

Intrinsic and extrinsic semiconductors, Fermi level and carrier- concentration, Effect of temperature on Fermi level. Mobility of charge carriers and effect of temperature on mobility.

Unit V: Optics (8)

Interference- Introduction and examples, Youngs; double slit experiment, Diffraction – Types, Single Slit, Double Slit, Diffraction Grating.

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Reference Books:

1. Engineering Physics by H.K. Malik and Singh,
2. Electrodynamics by David.J.Griffiths
3. Quantum Mechanics by Aruldhas.
4. Solid State Physics by C. Kittel.

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PH1701 **Engineering Physics Laboratory**

L-T-P-C
0-0-3-1.5

List of experiments:

1. Photoelectric effect
2. Hall effect
3. Ultrasonic Interferometer
4. Melde's Experiment
5. Four probe Method
6. Frank hertz Experiment
7. See beck and Peltier effect
8. Solar cell
9. Couple pendulum

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HS1101 **ENGLISH**

Externals: 60 Marks **L-T-P-C**
Internals: 40 Marks **2-0-0-2**

INTRODUCTION

In view of the growing importance of English as a tool for global communication and the consequent emphasis on training students to acquire language skills, the syllabus of English has been designed to develop linguistic, communicative and critical thinking competencies of Engineering students. In English classes, the focus should be on the skills development in the areas of vocabulary, grammar, reading and writing. For this, the teachers should use the prescribed text for detailed study. The students should be encouraged to read the texts leading to reading comprehension and different passages may be given for practice in the class. The time should be utilized for working out the exercises given after each excerpt, and also for supplementing the exercises with authentic materials of a similar kind, for example, newspaper articles, advertisements, promotional material etc. The focus in this syllabus is on skill development, fostering ideas and practice of language skills in various contexts and cultures.

Learning Objectives: The course will help to -

a. Improve the language proficiency of students in English with an emphasis on Vocabulary, Grammar, Reading and Writing skills.

b. Equip students to study academic subjects more effectively and critically using the theoretical and practical components of English syllabus.

c. Develop study skills and communication skills in formal and informal situations.

Course Outcomes: Students should be able to

1. Use English Language effectively in spoken and written forms.
2. Comprehend the given texts and respond appropriately.
3. Communicate confidently in various contexts and different cultures.
4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

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UNIT –I: The Raman Effect' from the prescribed textbook 'English for Engineers' published by Cambridge University Press (6)

Vocabulary Building: The Concept of Word Formation --The Use of Prefixes and Suffixes. **Grammar:** Identifying Common Errors in Writing with Reference to Articles and Prepositions. **Reading:** Reading and Its Importance- Techniques for Effective Reading. **Basic Writing Skills:** Sentence Structures -Use of Phrases and Clauses in Sentences Importance of Proper Punctuation- Techniques for writing precisely – **Paragraph writing** – Types, Structures and Features of a Paragraph - Creating Coherence-Organizing Principles of Paragraphs in Documents.

UNIT –II: Ancient Architecture in India' from the prescribed textbook 'English for Engineers' published by Cambridge University Press (6)

Vocabulary: Synonyms and Antonyms.

Grammar: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement.

Reading: Improving Comprehension Skills – Techniques for Good Comprehension

Writing: Format of a Formal Letter-**Writing Formal Letters** E.g.., Letter of Complaint, Letter of Requisition, Job Application with Resume.

UNIT –III: Blue Jeans from the prescribed textbook 'English for Engineers' published by Cambridge University Press (6)

Vocabulary: Acquaintance with Prefixes and Suffixes from Foreign Languages in English to form Derivatives-Words from Foreign Languages and their Use in English. **Grammar:** Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.

Reading: Sub-skills of Reading- Skimming and Scanning **Writing**: Nature and Style of Sensible Writing- **Defining- Describing** Objects, Places and Events – **Classifying**- Providing Examples or Evidence.

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UNIT –IV: What Should You Be Eating' from the prescribed textbook 'English for Engineers' published by Cambridge University Press (6)

Vocabulary: Standard Abbreviations in English

Grammar: Redundancies and Clichés in Oral and Written Communication.

Reading: Comprehension**-** Intensive Reading and Extensive Reading

Writing: Writing Practices--Writing Introduction and Conclusion - Essay Writing-Précis Writing.

UNIT –V: How a Chinese Billionaire Built Her Fortune' from the prescribed textbook 'English for Engineers' published by Cambridge University Press (6)

Vocabulary: Technical Vocabulary and their usage

Grammar: Common Errors in English

Reading: Reading Comprehension-Exercises for Practice

Writing: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report.

Prescribed Textbook:

1. Sudarshana, N.P. and Savitha, C. (2018). English for Engineers. Cambridge University Press.

References:

1. Swan, M. (2016). Practical English Usage. Oxford University Press.

2. Kumar, S and Lata, P.(2018). Communication Skills. Oxford University Press.

3. Wood, F.T. (2007).Remedial English Grammar. Macmillan.

4. Zinsser, William. (2001). On Writing Well. Harper Resource Book.

5. Hamp-Lyons, L. (2006).Study Writing. Cambridge University Press.

6. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.

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HS1701 **ENGLISH LANGUAGE LAB**

Orals (Written): 50Marks **L-T-P-C**
Written (Externals): 50Marks **0-0-2-1**

Course Objectives:

- To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning.
- To sensitize students to the nuances of English speech sounds, word accent, intonation and rhythm.
- To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
- To improve the fluency of students in spoken English and neutralize their mother tongue influence
- To train students to use language appropriately for public speaking and interviews

Syllabus

Listening Skills:

Objectives:

- To enable students develop their listening skills so that they may appreciate its role in the LSRW skills approach to language and improve their pronunciation.
- To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.
- Listening for general content
- Listening to fill up information
- Intensive listening for specific information

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Speaking Skills:

Objectives:

- To involve students in speaking activities in various contexts
- To enable students express themselves fluently and appropriately in social and professional contexts
- Oral practice: Just A Minute (JAM) Sessions Describing objects/situations/people
- Role play – Individual/Group activities

Course Outcomes:

Students will be able to attain

- Better understanding of nuances of English language through audio- visual experience and group activities
- Neutralization of accent for intelligibility

Speaking skills with clarity and confidence which in turn enhances their employability skills.

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BS1101

ENVIRONMENTAL SCIENCE

L-T-P-C
2-0-0-0

UNIT 1: MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL

Definition, scope and importance, need for public awareness.

UNIT 2: NATURAL RESOURCES:

Renewable and non-renewable resources: Natural resources and associated problems.

a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.

b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.

c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

d) Food resources: World food problems, changes caused by agriculture and over-grazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.

e) Energy resources: Growing energy needs, renewable and non renewable energy sources, use of alternate energy sources.

f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification.

- ✓ Role of an individual in conservation of natural resources.
- ✓ Equitable use of resources for sustainable lifestyles.

UNIT 3: ECOSYSTEMS & BIODIVERSITY

Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs and ecological pyramids.

Introduction, types, characteristic features, structure and function of the following ecosystems:-

- Forest ecosystem, b. Grassland ecosystem, c. Desert ecosystem, d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

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- b. Biodiversity- Definition : genetic, species and ecosystem diversity. Bio geographical classification of India Value of bio diversity: consumptive use, productive use, social, ethical, aesthetic and option values.
- c. Biodiversity at global, National and local levels. India as a mega-diversity nation and Hot-sports of biodiversity.
- d. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT 4: ENVIRONMENTAL POLLUTION

Definition, Cause, effects and control measures of :- Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards

- Solid waste Management: Causes, effect s and control measures of urban and industrial wastes.
- Role of an individual in prevention of pollution
- Pollution case studies.
- Disaster management: floods, earthquake, cyclone and landslides.
- Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies.
- Environment Protection Act., Air (Prevention and Control of Pollution) Act. Water Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act.

UNIT 5 : SOCIAL ISSUES & THE ENVIRONMENT

Human Rights. Value Education. HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.

Field work: Visit to a local area to document t environmental assets river/ forest/grassland/hill/mountain Visit to a local polluted site-Urban/Rural/Industrial/Agricultural. Study of common plants, insects, birds. Study of simple ecosystems-pond, river, hill slopes, etc.

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REFERENCES:

a). Agarwal, K.C. 2001 Environmental Biology, Nidi Publ. Ltd. Bikaner.

b). Bharucha Erach, The Biodiversity of India, Mapin Publishing Pvt. Ltd., Ahmedabad- 380 013, India, Email:mapin@icenet.net (R)

c). Brunner R.C., 1989, Hazardous Waste Incineration, McGraw Hill Inc. 480p

d) Clark R.S., Marine Pollution, Clanderson Press Oxford (TB)

e). Cunningham, W.P. Cooper, T.H. Gorhan i, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumabai, 1196p .

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ENGINEERING GRAPHICS

L-T-P-C
0-1-4-3

Course Objectives:

- To introduce the students to the "Universal Language of Engineers" for effective communication through drawing.
- To understand the basic concepts of drawing through modern techniques.
- To impart knowledge about standard principles of projection of objects.
- To provide the visual aspects of Engineering drawing using Auto-CAD.

UNIT-I: Introduction to Engineering Drawing (6)

Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, types of lines and Dimensioning.

Geometrical constructions: Construction of regular polygons. Introducion to scales, Conic sections and Cycloidal curves (Conventional)

UNIT-II: Orthographic projections, Projections of Points and Lines (8)

Orthographic projections: Principles of Orthographic Projections

Projections of Points: Projections of Points placed in different quadrants

Projection of lines: lines parallel and inclined to both the planes (Determination of true lengths and true inclinations and traces) (Conventional)

UNIT-III: Fundamentals of Auto CAD and Projection of planes (6)

Fundamentals of Auto CAD: Introduction to Auto-CAD, DRAW tools, MODIFY tools, TEXT, DIMENSION, PROPERTIES

Projection of planes: Planes inclined to both the reference planes (Auto-CAD)

UNIT-IV: Projection of Solids and Projection of sectioned solids (9)

Projection of Solids: Projection of solids whose axis is parallel to one of the reference planes and inclined to the other plane, axis inclined to both the planes (Auto-CAD)

Projection of sectioned solids: Sectioning of simple solids like prism, pyramid, cylinder and cone in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section.
(Auto-CAD)

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UNIT-V: Development of surfaces, Isometric Projections and Perspective projections (9)

Development of surfaces: Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone (Auto-CAD)

Isometric Projections: Principles of Isometric projection – Isometric Scale, Isometric Views of planes and simple solids (Auto-CAD)

Perspective projections: Basic concepts of perspective views.

Course Outcomes:

At the end of the course, the student will be able to

- Use Engineering principles and techniques to understand and interpret engineering drawings.
- Understand the concepts of Auto-CAD.
- Draw orthographic projections of lines, planes and solids using Auto-CAD.
- Use the techniques, skills and modern engineering tools necessary for engineering practices.

Text/Reference Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House

2. Gopalakrishna K.R., “Engineering Drawing” (Vol. I&II combined), Subhas Stores, Bangalore, 2007.

3. Shah, M.B. &Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education

4. Venugopal K. and Prabhu Raja V., “Engineering Graphics”, New Age publications

5. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication

6. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers

7. (Corresponding set of) CAD Software Theory and User Manuals

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EE1101 NETWORK THEORY-1

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives:

- To introduce the basic concepts of circuit analysis this is the foundation for all subjects of the Electrical Engineering discipline.
- To introduce the various techniques/tools used in the transient and steady-state response of electrical circuits.

Course Outcomes: At the end of this course, students will demonstrate the ability to

- Evaluate steady state behavior of single port networks for DC and AC excitations.
- Apply network theorems for the analysis of electrical circuits.
- Analyze and solve 3-phase circuits

UNIT I : DC Circuit Analysis (12 hours)

DC Circuits: Electrical circuit elements (R, L and C), voltage and current sources, Kirchhoff current and voltage laws, analysis of simple circuits with dc excitation, Network reduction techniques (Series, Parallel connection, Star-delta transformation), Analysis with dependent current and voltage sources. Nodal and Mesh Analysis.

UNIT II: Sinusoidal steady state analysis (10 Hours)

Representation of sine function as rotating phasor, phasor diagrams, impedances and admittances, AC circuit analysis, effective or RMS values, reactive power, apparent power, power factor, average power and complex power. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel).

UNIT III: Network theorems (10 Hours)

DC Network Theorems: Superposition theorem, Thevenin’s theorem, Norton’s theorem, Maximum power transfer theorem, Reciprocity theorem, Compensation theorem, Tellegen’s Theorem, Milliman’s Theorem.

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AC Network Theorems: Superposition theorem, Thevenin's theorem, Norton theorem, Maximum power transfer theorem.

UNIT IV: Resonance (6 Hours)

Resonance – Series resonance and Parallel resonance circuits, concept of bandwidth and Q factor, Duality & Dual Networks.

UNIT V: Three Phase circuits (10 Hours)

Three phase EMF generation, Phase Sequence, Star and Delta Connections, Three phase balanced circuits, line and phase quantities, voltage and current relations in star and delta connections, solution of three phase circuits, Analysis of Balanced Three Phase Circuits, , Three phase four wire circuits, Analysis of Unbalanced Three Phase Circuits.

Text Books:

1. M. E. Van Valkenburg, " Network Analysis" , Prentice Hall, 2006.
2. D. Roy Choudhury, " Networks and Systems", New Age International Publications, 1998.
3. W. H. Hayt and J. E. Kemmerly, " Engineering Circuit Analysis" , McGraw Hill Education, 2013.
4. Network Theory by N.C.Jagan & C.Lakshminarayana, B.S. Publications.
5. Network Theory by Sudhakar, Shyam Mohan Palli, TMH.

Reference Books:

1. C. K. Alexander and M. N. O. Sadiku, " Electric Circuits", McGraw Hill Education, 2004.
2. K. V. V. Murthy and M. S. Kamath, " Basic Circuit Analysis" , Jaico Publishers, 1999.
3. E. Hughes, "Electrical and Electronics Technology", Pearson, 2010.

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FIRST YEAR (E1) SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE1201	Network Theory-II	PCC	3	0	0	3	3
2	EE1801	Network Theory Lab	ESC	0	0	3	3	1.5
3	MA1202	Differential equations andLaplace Transforms	BSC	3	1	0	4	4
4	CY1201	Engineering Chemistry	BSC	3	0	0	3	3
5	CY1801	Engineering Chemistry Lab	BSC	0	0	3	3	1.5
6	CS1202	Programming for ProblemSolving	ESC	3	0	0	3	3
7	CS1802	Programming for ProblemSolving Lab	ESC	0	0	3	3	1.5
8	ME1802	Engineering Workshop	ESC	0	1	2	3	2
9	BM1205	Constitution of India	MC	2	0	0	2	0
10	EC1203	Analog Electronic Circuits-I	ESC	3	0	0	3	3
Total				17	2	11	30	22.5

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EC1203 ANALOG ELECTRONIC CIRCUITS-I

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives:

- To introduce the semiconductor devices like Diode, BJT, MOSFET and Their applications
- introduce components such as diodes, BJTs and FETs their Switching characteristics,applications
- Learn the concepts of high frequency analysis of transistors.
- Learn the concept of Rectification and some wave shaping circuits.

Course Outcomes:

- On completion of this course, the students will be able to
- Analyze conduction in semiconductors and estimate the diode parameters from its characteristics.
- Examine the performance characteristics of rectifiers with and without filters.
- Design the biasing circuits; compare the various configurations of BJT
- Design of FET biasing circuits.

UNIT-I: Review of Semiconductor Physics:(8 Hours)

Conductivity of a semiconductor, carrier concentrations in an intrinsic semiconductor, donor and acceptor impurities, charge densities in a semiconductor, Fermi level in a semiconductor with impurities, continuity equation, the Hall Effect.

UNIT-II: Diode Characteristics and its Applications:**(10 Hours)**

Qualitative theory of P-N junction diode, current equation, I-V characteristics, temperature dependent parameters, diode resistance, transition capacitance, diffusion capacitance, Zener diode.

Rectifiers: Diode as a switch, half wave and Full wave (Center tapped and Bridge type), Peak Inverse voltage, form factor, ripple factor, Transformer Utilization factor and Efficiency, Wave shaping circuits: Clamping, clipping circuits and slicer.

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NIT-III: BJT CIRCUITS:(12 Hours)

Construction and working of BJT, I-V characteristics, BJT as a switch. Transistor configurations, Early Effect, transistor Biasing, thermal runaway and thermal stability the operating point, Biasing Techniques, bias compensation. BJT as amplifier. Small signal equivalent circuits.

UNIT IV: FIELD EFFECT TRANSISTORS: (6 Hours)

Construction and working of FET, I-V Characteristics, Types of FETS, MOSFET as a Switch and as a amplifier.

UNIT V: MOSFET CONFIGURATIONS AND SMALL SIGNAL ANALYSIS(12 Hours)

MOSFET configurations, biasing circuits, small signal equivalent circuits - gain, input and output impedances, trans-conductance, high frequency equivalent circuit.

TEXT BOOKS:

1. Robert L Boylested and Louis Nashelsky, Electronic Devices and Circuit Theory, 10th ed., New Delhi: Pearson Ind. Pvt. Ltd., 2009
2. Jacob Millman, Christos C Halkias & Satyabrata JIT, Electronic Devices and Circuits, New Delhi: Tata McGraw Hill Education (INDIA) Private Ltd, 2007.

REFERENCE BOOKS:

1. Electronic Devices Conventional and current version -Thomas L. Floyd 2015, pearson.
2. Linear Integrated Circuits 4th Edition By Dr. Roy Choudhury, shail B.Jain.
3. P.R. Gray, R.G Meyer and S. Lewis, "Analysis and Design of Analog Integrated Circuits", John Wiley &Sons,2001

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MA1202 **Differential Equations and Laplace Transforms**

Externals: 60 marks
Internals: 40 Marks

L	T	P	C
3	1	0	4

Course Objectives:

- Methods of solving the differential equations of first and higher order.
- To Solve the Differential & integral equations using Laplace Transform
- To understand the Applications of Laplace Transforms

Course Outcomes: At the end of the course student will be able to

- Solve first order linear differential equations and special non linear first order equations like Bernouli , Riccati&Clairaut's equations
- Use shift theorems to compute the Laplace transform and inverse Laplace transform
- Use the Laplace transform to compute solutions of equations involving impulse functions

UNIT-I: Ordinary Differential Equations of first order

Exact first order differential equation, finding integrating factors, linear differential equations, Bernoulli's , Riccati , Clairaut's differential equations, finding orthogonal trajectory of family of curves, Newton's Law of Cooling, Law of Natural growth or decay.

UNIT-II: Ordinary Differential Equations of higher order

Linear dependence and independence of functions, Wronskian of n- functions to determine Linear Independence and dependence of functions, Solutions of Second and higher order differential equations (homogeneous & non-homogeneous) with constant coefficients, Method of variation of parameters, Euler-Cauchy equation.

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UNIT-III: Laplace Transform –I:

Definition of Laplace Transform, linearity property, conditions for existence of Laplace Transform. First and second shifting properties, Laplace Transform of derivatives and integrals, unit step functions, Dirac delta-function, error function. Differentiation and integration of transforms, convolution theorem.

UNIT-IV Inverse Laplace Transforms

Finding Inverse Laplace Transform using various methods, Evaluation of integrals by Laplace Transform. Solving initial and boundary value problems, Differential Equations & Partial differential equations, Integral Equations using Laplace Transforms.

UNIT-V: Integral Calculus

Convergence of improper integrals, tests of convergence, Beta and Gamma functionselementary properties, differentiation under integral sign, differentiation of integrals with variable limits.Leibnitz rule. Rectification, double and triple integrals, computations of surface and volumes, change of variables in double integrals - Jacobians of transformations, integrals dependent on parameters – applications.

Text Books:

1. Advanced Engineering Mathematics (3rd Edition) by R. K. Jain and S. R. K. Iyengar, Narosa Publishing House, New Delhi

References Books:

1. Advanced Engineering Mathematics (8th Edition) by Erwin Kreyszig, Wiley-India.
2. Dr.M.D.Raisinghania,Ordinary & Partialdifferentialequations,S.CHAND,17thEdition2014.

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EE1201 **NETWORK THEORY-II**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- To introduce the various techniques used in the transient and steady-state response of electrical circuits.
- To understand the first order and second order differential equations.
- To learn about two port networks and network topology.
- The emphasis of this course is laid on the basic analysis of circuits which includes Circuit concepts, magnetic circuits.

Course Outcomes:

At the end of this course, students will demonstrate the ability to

- Evaluate transient behavior of single port networks for DC and AC excitations.
- Examine behavior of linear circuits using Laplace transform and transfer functions of single port and two port networks.
- Understand the concept of magnetic circuits.
- Analyze two port circuit behaviour.

UNIT I: Solution of First and Second order networks (10 Hours)

Solution of first and second order differential equations for Series and parallel R-L, R-C, R-L-C circuits, initial and final conditions in network elements, forced and free response, time constants, steady state and transient state response. Transient Response of R-L, R-C, R-L-C Series Circuits for Sinusoidal Excitation.

UNIT II: Electrical Circuit Analysis Using Laplace Transforms (10 Hours)

Review of Laplace Transform, Analysis of electrical circuits using Laplace Transform for standard inputs, inverse Laplace transform, and transformed network with initial conditions. Transfer function representation. Poles and Zeros. Frequency response (magnitude and phase plots).

UNIT III: Two Port Network (10 Hours)

Open circuit impedance, short-circuit admittance, Transmission, Hybrid parameters & inter-relationships, series, parallel and cascade connection of two port networks, system

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function, impedance and admittance functions.

UNIT IV: Magnetic Circuits:(8 Hours)

Faraday's Laws of Electromagnetic Induction, Concept of Self and Mutual Inductance, Mutual coupled circuits, Coefficient of Coupling, Dot Convention, Ideal Transformer, Composite Magnetic Circuit-Analysis of Series and Parallel Magnetic Circuits.

UNIT V: Network topology and Network Synthesis (10 Hours)

Network Topology: Definitions – Graph, Tree, chord, Basic Cut-set, Basic Tie set and incident Matrices for Planar Networks, Loop and Nodal Analysis of Networks with Dependent & Independent Voltage and Current Sources, Duality & Dual Networks.

Network synthesis: Network synthesis of driving point functions, Positive real functions Hurwitz polynomials, Realization of passive RL, RC and LC networks using Foster and Caner forms.

Text Books:

1. C. K. Alexander and M. N. O. Sadiku, "Electric Circuits", McGraw Hill Education, 2004.
2. Chakravarthy A.,Circuit Theory,Dhanpat Rai & Co., First Edition, 1999.

Reference Books:

1. M. E. Van Valkenburg, "Network Analysis", Prentice Hall, 2006.
2. D. Roy Choudhury, "Networks and Systems", New Age International Publications, 1998.
3. W. H. Hayt and J. E. Kemmerly, "Engineering Circuit Analysis", McGraw Hill Education, 2013.
4. Network Theory by N.C. Jagan & C. Lakshminarayana, B.S. Publications.
5. Network Theory by Sudhakar, Shyam Mohan Palli, TMH.

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EE1801 **NETWORK THEORY LAB**

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0- 0 - 3 – 1.5**

Course Objective:

- To expose the students to the concepts of electrical and electronics circuits and give them experimental skills.

Course Outcomes: Upon completion of this course

- The student will be able to perform experiments to verify network theorems.
- Understand the usage of common electrical measuring instruments.
- The student will be able to perform experiments to study transient and steady state behavior of electrical circuits for DC and Sinusoidal excitation.
- The student will be able to perform experiments to determine the two port network parameters.

List of Experiments:

1. Introduction Lab
   i. Basic safety precautions. Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.
   ii. Verification of Ohm's Law, KCL and KVL
2. Measuring the steady-state and transient time-response of R-L and R-C to a step change in voltage (transient may be observed on a storage oscilloscope).
3. Measuring the steady-state and transient time-response of R-L-C circuits to a step change in voltage (transient may be observed on a storage oscilloscope).
4. Sinusoidal steady state response of R-L, and R-C circuits – impedance calculation and verification. Observation of phase differences between current and voltage.
5. Resonance in R-L-C circuits.
6. Verification of Superposition theorem and Reciprocity theorem
7. Verification of Thevenin's theorem and Maximum power transfer theorem
8. Determination of 2-port parameters for a given network: Z,Y, ABCD parameters

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CY1201 **Engineering Chemistry**

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I: Electrochemistry (12 Hours)

Introduction to electrochemistry: Galvanic cell (Daniel cell), Nernst equation. Types of electrodes: metal-metal ion electrodes, metal-insoluble salt-anion electrodes, calomel electrode, gas-ion electrodes, hydrogen and chlorine electrodes, oxidation-reduction electrodes (quinhydrone electrode), amalgam electrodes and ion exchange electrode (glass electrode). EMF and applications of EMF: determination of pH of the solution, potentiometric titrations, Classificationof commercial cells - primary cells (dry cell) and secondary cells (Lithium ion battery, Pb-acid storage battery). Fuel cells: H_2-O_2 fuel cell.

UNIT - II: Corrosion and Water Treatment. (10 Hours)

Dry and wet corrosion and their mechanisms. Pilling - Bedworth Rule. Types of Corrosion: galvanic corrosion, concentration cell corrosion, pitting corrosion and stress corrosion. Factors influencing the rate of corrosion: Temperature, pH and dissolved oxygen. Corrosion Prevention methods: Cathodic protection – Sacrificial Anodic method and Impressed current method. Metalliccoatings: galvanization and tinning methods.

Water: Hardness of water, Degrees of hardness. Calculation of hardness by EDTA method. Disadvantages of hard water in boilers: priming, foaming, scales, sludges and caustic embrittlement. Treatment of boiler feed water: Zeolite process, Ion exchange process.

UNIT - III: Energy sources (10 Hours)

Introduction. Definition and classification of fuels. Calorific value of a fuel, Characteristics of a good fuel. Coal, types of Coal. Analysis of Coal: Proximate and Ultimate analysis. Bomb Calorimeter and Junker's gas Calorimeter. Problems on calculation of calorific value. Liquid fuels Introduction .Synthetic Petrol: Fisher Tropsch process. Introduction to Bio-fuels: Bio-diesel, Bio-gas.

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Unit IV: Chemical Kinetics (8 Hours)

Introduction to rate of reaction and rate constant determination. Factors influencing rate of reaction. Complex reactions: definition and classification of complex reactions, definition of reversible reactions with examples, rate law derivation for reversible reactions. Consecutive reactions: definition, rate law derivation and examples of consecutive reactions. Parallel reactions: definition, rate law derivation and examples of parallel reactions. Steady-state approximation: introduction, kinetic rate law derivation by applying steady state approximation in case of the oxidation of NO and pyrolysis of methane.

UNIT - V: Nanochemistry (10 Hours)

Introduction to nanomaterials, classification: Carbon based nanomaterials, metallic nanoparticles, metal oxide nanoparticles. Properties at nanoscale. Synthetic approaches: Top-Down (Photolithography, ball milling) and Bottom-Up (Sol-gel, Hydrothermal). Brief overview on characterization of nanomaterials: X-ray, SEM and TEM. Applications of nanomaterials.

Reference Books

1. Engineering Chemistry, Jain & Jain
2. Engineering Chemistry, Shashi Chawla
3. Chemistry for Engineers, B. K. Ambasta
4. Engineering Chemistry, H. C. Srivastava
5. Fundamentals of engineering Chemistry by Shikha Agarwal.

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CY1801 **ENGINEERING CHEMISTRY LAB**

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 3 – 1.5**

List of the experiments for Engineering Chemistry

1. Determination of the strength of weak acid (CH_3COOH) by pH metry.
2. Conductometric titration (strong acid (HCl) vs strong base (NaOH)).
3. Throwing power of Copper.
4. Estimation of alkalinity of water.
5. Determination of total hardness of water by complexometric method using EDTA.
6. Determination of the calorific value of fuel sample by using bomb calorimeter.
7. Preparation of bio-diesel from palm oil by trans esterification method.
8. The rate constant and order of the reaction of the hydrolysis of an ester catalyzed by an acid(dil.HCl).
9. Preparation of Nano particle (ZnO).

Reference books:

1) Essentials of experimental engineering chemistry by Shashi chawla.
2) Practical chemistry by Dr.O.P.Pandey, S.Chand publication.
3) A textbook of engineering chemistry by Shashi chawla.
4) College practical chemistry by VK Ahluwalia.
5) Practical engineering chemistry by K. Mukkanti.
6) Laboratory manual by R. Kulakarni, Adil.

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CS1202

Programming for Problem Solving

Internal Marks 40
External Marks 40

L-T-C-P
3-0-0-3

Course Objectives

- To learn the fundamentals of computers.
- To understand the various steps in program development.
- To learn the syntax and semantics of the C programming language.
- To learn the usage of structured programming approaches in solving problems.

Course Outcomes

1. To write algorithms and to draw flowcharts for solving problems.
2. To convert the algorithms/flowcharts to C programs.
3. To code and test a given logic in the C programming language.
4. To decompose a problem into functions and to develop modular reusable code.
5. To use arrays, pointers, strings and structures to write C programs.
6. Searching and sorting problems.

UNIT-I: Introduction to Programming

Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.)

Representation of Algorithm - Algorithms for finding roots of a quadratic equations, finding minimum and maximum numbers of a given set, finding if a number is prime number Flowchart/Pseudo code with examples, Program design and structured programming.

Introduction to C Programming Language: variables (with data types and space requirements), Syntax and Logical Errors in compilation, object and executable code, Operators, expressions and precedence, Expression evaluation, type conversion, The main method and command line arguments, Bitwise operations: Bitwise AND, OR, XOR and NOT operators.

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UNIT-III: Strings, Structures, Pointers

Strings: Introduction to strings, handling strings as array of characters, basic string functions available in C (strlen, strcat, strcpy, strstr etc.), arrays of strings

Structures: Defining structures, initializing structures, unions, Array of structures

Pointers: Idea of pointers, Defining pointers, Pointers to Arrays and Structures.

UNIT-IV

Functions, Recursion, Preprocessor, Storage classes and Dynamic Memory Allocation

Functions: Designing structured programs, declaring a function, Signature of a function, Parameters and return type of a function, passing parameters to functions, call by value, Passing arrays to functions, passing pointers to functions, idea of call by reference.

Recursion: Simple programs, such as Finding Factorial, Fibonacci series etc., Limitations of Recursive functions

Preprocessor: Commonly used Preprocessor commands like include, define, undef, if, ifdef, if and if Storage classes (auto, extern, static and register),

Dynamic memory allocation: Allocating and freeing memory.

UNIT-V: Files, Searching and Sorting

Files: Text and Binary files, Creating and Reading and writing files, Appending data to existing files. Basic searching in an array of elements (linear and binary search techniques), Basic algorithms to sort array elements (Bubble, Selection and Insertion sort).

Text Books

- .Jeri R. Hanly and Elliot B.Koffman, Problem solving and Program Design in C 7th Edition, Pearson
- B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)

References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression)
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
6. Herbert Schildt, C: The Complete Reference, McGraw Hill, 4th Edition
7. 7. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill

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CS1802 **Programming for Problem Solving Lab**

Internals: 40 Marks
Externals: 60 Marks

L-T-P-C
0-0-3-1.5

Course Objectives

- To work with an IDE to create, edit, compile, run and debug programs
- To analyze the various steps in program development.
- To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
- To develop modular, reusable and readable C Programs using the concepts like functions, arrays etc.
- To write programs using the Dynamic Memory Allocation concept.
- To create, read from and write to text and binary files

Course Outcomes

The candidate is expected to be able to:

1. Formulate the algorithms for simple problems
2. Translate given algorithms to a working and correct program
3. Correct syntax errors as reported by the compilers
4. Identify and correct logical errors encountered during execution
5. Represent and manipulate data with arrays, strings and structures
6. Use pointers of different types
7. Create, read and write to and from simple text and binary files
8. Modularize the code with functions so that they can be reused

Lab Experiments

Practice sessions:

A. Write a simple program that prints the results of all the operators available in C (including pre/ post increment, bitwise and/or/not , etc.). Read required operand values from standard input.
B. Write a simple program that converts one given data type to another using auto conversion and casting. Take the values from standard input.

Simple numeric problems:

A. Write a program for finding the max and min from the three numbers.

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B. Write the program for the simple, compound interest.
C. Write a program that declares Class awarded for a given percentage of marks, where mark <40%= Failed, 40% to <60% = Second class, 60% to <70%=First class, >= 70% = Distinction. Read percentage from standard input.
D. Write a program that prints a multiplication table for a given number and the number of rows in the table. For example, for a number 5 and rows = 3, the output should be:

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

E. Write a program that shows the binary equivalent of a given positive number between 0 to 255.

Expression Evaluation:

A. A building has 10 floors with a floor height of 3 meters each. A ball is dropped from the top of the building. Find the time taken by the ball to reach each floor. (Use the formula s = ut+(1/2)at^2 where u and a are the initial velocity in m/sec (= 0) and acceleration in m/sec^2 (= 9.8 m/s^2)).
B. Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement)
C. Write a program that finds if a given number is a prime number
D. Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome.
E. A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.
F. Write a C program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.
G. Write a C program to find the roots of a Quadratic equation.
H. Write a C program to calculate the following, where x is a fractional value. i. 1-x/2 +x^2/4-x^3/6.
I. Write a C program to read in two numbers, x and n, and then compute the sum of this geometric progression: 1+x+x^2+x^3+ … +x^n. For example: if n is 3 and x is 5, then the program computes 1+5+25+125.

A. Write C programs that use both recursive and non-recursive functions
To find the factorial of a given integer.

B. Write a program for reading elements using a pointer into an array and display the values using the array.
C. Write a program for display values reverse order from an array using a pointer.
D. Write a program through a pointer variable to sum of n elements from an array.

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Arrays, Pointers and Functions:

- **A.** Write a C program to find the minimum, maximum and average in an array of integers.
- **B.** Write a function to compute mean, variance, Standard Deviation, sorting of n elements in a single dimension array.
- **C.** Write a C program that uses functions to perform the following:
    - **a.** Addition of Two Matrices
    - **b.** Multiplication of Two Matrices
    - **c.** Transpose of a matrix with memory dynamically allocated for the new matrix as row and column counts may not be the same.
    - **d.** To find the GCD (greatest common divisor) of two given integers.
    - **e.** To find x^n.

Files:

- **A.** Write a C program to display the contents of a file to standard output device.
- **B.** Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.

Write a C program to count the number of times a character occurs in a text

- **A.** file. The file name and the character are supplied as command line arguments.
- **B.** Write a C program that does the following:
    - **a.** It should first create a binary file and store 10 integers, where the file name and 10 values are given in the command line. (hint: convert the strings using atoi function)

Now the program asks for an index and a value from the user and the value

- **a.** at that index should be changed to the new value in the file. (hint: use fseek function)
- **b.** The program should then read all 10 values and print them back.
- **C.** Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Strings:

Write a C program to convert

- **A.** a Roman numeral ranging from I to L to its decimal equivalent.
- **B.** Write a C program that converts a number ranging from 1 to 50 to Roman equivalent
- **C.** Write a C program that uses functions to perform the following operations:
    - **a.** To insert a sub-string into a given main string from a given position.

To delete n Characters from

- **A.** a given position in a given string.
- **B.** Write a C program to determine if the given string is a palindrome or not (Spelled same in both directions with or without a meaning like madam, civic, noon, abcba, etc.)
- **C.** Write a C program that displays the position of a character ch in the string S or – 1 if S doesn‘t contain ch.
- **D.** Write a C program to count the lines, words and characters in a given text.

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E. between finding the smallest, largest, sum, or average. The menu and all the choices are to be functions. Use a switch statement to determine what action to take. Display an error message if an invalid choice is entered.
F. Write a C program to construct

Miscellaneous:

a. Write a menu driven C program that allows a user to enter n numbers and then choose

A. a pyramid of numbers as follows:

1	*	1	1	*
1 2	* *	2 3	2 2	* *
1 2 3	* * *	4 5 6	3 3 3	* * *
			4 4 4 4	* *

Sorting and Searching:

A. Write a C program that uses non recursive function to search for a Key value in a given list of integers using linear search method.
B. Write a C program that uses non recursive function to search for a Key value in a given sorted list of integers using binary search method.
  f. Write a C program that implements the

A. Bubble sort method to sort a given list of integers in ascending order.
B. Write a C program that sorts the given array of integers using selection sort in descending order
C. Write a C program that sorts the given array of integers using insertion sort in ascending order
D. Write a C program that sorts a given array of names

*

References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, PHI
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression)
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
6. Herbert Schildt, C: The Complete Reference, Mc Graw Hill, 4th Edition
7. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill

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ME1802

Engineering Workshop

Internals: 40 Marks L - T - P - C
Externals: 60 Marks 0 - 1 - 2 - 2

Course Outcomes: Upon completion of this laboratory course

- Students will be able to fabricate components with their own hands.

List of Experiments:

1. **Fitting –** To produce a Step Fit on the given work piece.
   - To produce a V Fit on the given work piece.
2. **Carpentry** – To produce a Half lap joint on the given wooden work part.
   - To produce a Dove tail joint on the given wooden work part.
3. **House Wiring** – To perform and understand the Series and Parallel wiring connections.
   - To perform and understand Staircase and Godown wiring connections.
4. **Tin Smithy** – To produce a Tray from the given sheet metal.
   - To produce a Cylinder from the given sheet metal.
5. **Welding** – To practice formation of a Bead on the given work piece.
   - To perform a Butt and a Lap joint on the given work piece.
6. **Foundry** – To prepare a Mold cavity using a Single piece pattern.
   - To prepare a Mold cavity using a Split piece pattern.
7. **Machining** – To perform a Plain turning operation, Facing operation on the given work piece.
   - To perform a Step and a Taper turning operation on the given work piece.
8. **Plastic molding** – Demonstration
9. **WIRE EDM, CNC, 3D Printer** - Demonstration

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BM1205

CONSTITUTION OF INDIA

(COI)

Internal Marks 40 **L-T-P-C**
External Marks 60 **2-0-0-0**

Course Out comes: At the end of the course the student will be able to

CO1: Understand the formation and principles of Indian Constitution.

CO2: Understand Fundamental Rights and its implications in life

CO3: Understand Fundamental Duties of Individual toward country and society

CO4: Understand Directive principles to govern the policy formation

CO5: Understand the Way of running the Government and basic Governance

UNIT-I: Introduction to Indian Constitution

- Meaning of the term Constitution
- Historical background of Indian constitution
- Making of Indian constitution
- Constituent Assembly

UNIT-II: Features of Indian Constitution

- Preamble of the Constitution , Importance, Scope, Relevance
- The Salient Features of Indian Constitution , Importance, Scope, Relevance

UNIT-III: Fundamental Rights

- Fundamental Rights
- Importance and scope of fundamental rights
- Categorization of Fundamental Rights

UNIT-IV: Fundamental Duties & The Directive Principles of State Policy

- Fundamental Duties
- Importance and scope of fundamental Duties
- The Directive Principles of State Policy - Importance, Scope, Relevance

UNIT-V: Union/Central Government

- Union Government
- Union Legislature (Parliament)
- Lok Sabha and Rajya Sabha (with Powers and Functions)
- Union Executive

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- President of India (with Powers and Functions)
- Prime Minister of India (with Powers and Functions)

TextBooks:

1. ‘Indian Polity’ by Laxmikanth
2. ‘Indian Administration’ by Subhash Kashyap
3. ‘Indian Constitution’ by D.D. Basu
4. ‘Indian Administration’ by Avasti and Avasti

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SECOND YEAR (E2) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE2101	Electrical Machines-I	PCC	3	1	0	4	4
2	EC2104	Analog Electronic Circuits-II	ESC	3	0	0	3	3
3	EC2103	Electromagnetic Fields	PCC	3	1	0	4	4
4	EC2701	Analog Electronic Circuits Lab	ESC	0	0	2	2	1
5	ME2105	Engineering Mechanics	ESC	3	1	0	4	4
6	MA2105	Vector Calculus and Complex analysis	BSC	3	1	0	4	4
7	EE2701	Electrical Machines-I Lab	PCC	0	0	3	3	1.5
8	CS2301	Data Structures **(Open Elective-I)**	OEC	3	0	0	3	3
Total				15	4	5	24	24.5

Electrical and Electronics Engineering

MA2105 VECTOR CALCULUS AND COMPLEX ANALYSIS

Internal Marks 40 **L-T-P-C**
External Marks 60 **3-1-0-4**

Course Objectives:

- The basic properties of vector valued functions and their applications to line, surface and volume integrals
- To Introduce Functions of Complex variables, Power series, Bilinear Transformations & Conformal Mapping.
- To learn various numerical methods for solving Differential equations.

Course outcomes: After learning the concepts of this paper the student must be able to

- Explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics.
- Recognize the statements of Green's theorem, Divergence Theorem Stokes' Theorem and understand how they are generalizations of the Fundamental Theorem of Calculus.
- Determine continuity/differentiability/analyticity of a function and find the derivative of a function
- Evaluate a contour integral using parameterization, fundamental theorem of calculus and Cauchy's integral formula
- Compute the residue of a function and use the residue theory to evaluate a contour integral or an integral over the real line
- Students will be familiar with numerical solutions of ordinary differential equations

UNIT-I: Vector Differentiation (10 Hours)

Vector point functions and scalar point functions. Gradient, Divergence and Curl. Directional derivatives, Tangent plane and normal line. Vector Identities. Scalar potential functions. Solenoidal and Irrotational vectors.

UNIT-II: Vector Integration (10 Hours)

Line Integration, Work done by a moving particle, Surface and Volume Integrals. Theorems of Green's theorem, Gauss Divergence theorem and Stoke's theorem (without proofs) and their applications in evaluation line,surface and volume integrals.

UNIT-III: Complex Variable – Differentiation (10 Hours)

Functions of complex variables. Limit continuity and differentiability of complex variable function. Elementary functions (exponential, trigonometric, logarithmic) . analytic function.

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Cauchy-Riemann equations. Harmonic functions- Harmonic conjugate. Conformal mappings, Mobious transformations and their properties.

UNIT-IV: Complex Variable-Integration (10 Hours)

Contour integrals, Cauchy-Goursat theorem (without proof), and Cauchy Integral formula (without Proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, Zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem(Without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour

UNIT-V: Numerical Solution of Ordinary Differential Equations (10 Hours)

Numerical solutions of IVP - Difference equations, stability, error and convergence analysis. Single step methods -Taylor series method, Euler method, Picard's method of successive approximation, Runge-Kutta Method(second order & fourth order) . Multi step methods. Newton's method for system of equations, stability, error and convergence analysis

TEXTBOOKS:

- Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8^{th} Edition,
- R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House.2008
- M.K Jain, S.R.K Iyengar, R.K Jain, computational methods for PDE, Wiley eastern 1994

REFERENCES:

- Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11^{th}Reprint, 2010.
- B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.
- Butcher, J.C 1987, the Numerical analysis of ordinary differential equations, Runge - Kutta and general linear methods. Wiley, New york.

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EC2104

ANALOG ELECTRONIC CIRCUITS-II

Externals: 60Marks **L-T-P-C**
Internals: 40Marks **3-0-0-3**

Course Objectives:

- To give understanding of various types of basic and feedback amplifier circuits such as smallsignal, cascaded, large signal and tuned amplifiers.
- To introduce the basic building blocks of linear integrated circuits.
- To introduce the concepts of waveform generation and introduce some special function ICs.
- To Understand the concept of sampling of feedback amplifiers

Course Outcomes:

- On completion of this course, the students will be able to…
- Design sinusoidal and non-sinusoidal oscillators.
- A thorough understanding the functioning of OP-AMP.
- Designs OP-AMP based circuits with linearintegrated circuits.
- Understanding of how Multi vibrator works.

UNIT-I: Multi-Stage and Power Amplifiers: (10Hours)

Direct coupled and RC Coupled multi-stage amplifiers; Differential Amplifiers, Power amplifiers - Class A, Class B, Class AB, Class C, Class D and Push pull Amplifier.

UNIT II: Feedback Amplifiers (10 Hours)

Concepts of feedback, Classification of feedback amplifiers, General characteristics of Negative feedback amplifiers, Effect of Feedback on Amplifier characteristics. Sampling and Mixing Concept: Voltage series, Voltage shunt, Current series and Current shunt Feedback configurations.

UNIT III: Operational Amplifiers: (10 Hours)

Ideal and Practical op-amps, its Characteristic (inputoffset current, inputoffset voltage, input bias current, Output offset voltage, slew rate, gain bandwidth product, Common mode gain, Differential mode gain ,Common Mode Rejection Ratio and Supply Voltage Rejection Ratio).

UNIT IV: APPLICATIONS OF OPERATIONAL AMPLIFIERTS (12 Hours)

Linear applications of op-amp: Idealized analysis of op-amp circuits. Inverting and non-inverting amplifier, differential amplifier, instrumentation amplifier, integrator, active filter, P, PI and PID controllers and lead/lag compensator using an op-amp.

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Nonlinear applications of op-amp:

Hysteretic Comparator, Zero crossing Detector, Square-wave and triangular-wave generators, Schmitt trigger, 555 timer and Multivibrators.

UNIT V: Oscillators (8 Hours)

Barkhausen stability criterion for Oscillations, RC type Oscillators-RC phase shift and Wien-bridge Oscillators, LC type Oscillators –Generalized analysis of LC Oscillators, Hartley, Clapp, Cristal and Colpitts Oscillators

TEXT BOOKS:

1. Robert L Boylested and Louis Nashelsky, Electronic Devices and Circuit Theory, 10th ed., New Delhi: Pearson Ind. Pvt. Ltd., 2009
2. Jacob Millman, Christos C Halkias & Satyabrata JIT, Electronic Devices and Circuits, New Delhi: Tata McGraw Hill Education (INDIA) Private Ltd, 2007.

REFERENCE BOOKS:

1. Electronic Devices Conventional and current version -Thomas L. Floyd 2015, pearson.
2. Linear Integrated Circuits 4th Edition By Dr. Roy Choudhury, shail B.Jain.
3. P.R. Gray, R.G Meyer and S. Lewis, “Analysis and Design of Analog Integrated Circuits”, John Wiley &Sons,2001

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EC2103

ELECTRO MAGNETIC FIELDS

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-1-0-4**

Course Objectives: To provide the basic knowledge

- Tofindelectricandmagneticfieldsforsymmetricalchargeandcurrentconfigurations.
- To deduce EM wave propagation in free space and in dielectric medium.
- Toanalyzeelectromagneticwavepropagationinguidingstructuresundervariousmatchingconditions
- To understand the power flow mechanism in the losssy and lossless transmission lines.

Course Outcomes: At the end of this course, students will demonstrate the ability to

- To understand the basic laws of electromagnetism.
- To obtain the electric and magnetic fields for simple configurations under static conditions.
- To analyze time-varying electric and magnetic fields.
- To understand Maxwell's equation in different forms and different media & understand the propagation of EM waves.

UNIT-I: Introduction of Vector Calculus(6 hours)

Vector algebra addition, subtraction, components of vectors, scalar and vector multiplication striple products, three orthogonal coordinate systems (rectangular, cylindrical and spherical).Vectorcalculusdifferentiation,partialdifferentiation,integration,vectoroperatordel,Gradient,divergence, and curl; integral theorems of vectors. Conversions of a vector from one coordinate system to another.

UNITII: Electrostatics(8Hrs)

Coulomb's Law, Electric Field Intensity due to different Charge Distributions, Electric Flux Density, Gauss Law and Applications, Electric Potential due to different Charge Distributions,RelationsBetweenEandV,EquipotentialSurfaces,EnergyDensityintheElectrostaticField.

UNITIII: Conductor, Dielectric and Boundary conditions(8Hrs)

Convection and Conduction Currents, current density, continuity Equation, conductor, Dielectricmaterialsandtheirproperties.Boundaryconditionsbetweenconductor-dielectrics, dielectric-dielectric and conductor-free space. Poisson's and Laplace's equations, General procedure for solving Poisson's and Laplace's equation. Capacitance calculations for different type of di-electric mediums, Capacitance of a two-wire line.

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UNITIV: Magneto statics (10Hrs)
BiotSavart's Law, Ampere's Circuital Law and Applications, Magnetic Flux Density, Magnetic Scalar and Vector Potentials, Forces due to Magnetic Fields, Magnetization in material, Magnetic torque and moments, Magnetic Energy, Self and mutual inductances.
Maxwell's Equations (Time Varying Fields): Faraday's Law, General Field Relation for Time Varying Electric And Magnetic Field, Maxwell's Equations in Different Forms, Conditions at a Boundary Surface : Dielectric-Dielectric and Dielectric-Conductor Interfaces. Poynting Vector and Poynting Theorem.

UNITV: Electromagnetic waves(10Hrs)
Wave Equations for Conducting Media, Uniform Plane Waves–Definition, Relations Between E&H, plane wave in good conductor, skin effect, skin depth.
Transmission Lines: Types, Parameters, Transmission Line Equations, Expressions for Characteristic Impedance, Input Impedance Relations, SC and OC Lines, Reflection Coefficient, UHF Lines as Circuit Elements : $\lambda/4$, $\lambda/2$, $\lambda/8$ Lines – Impedance Transformations, Single Stub Matching.

Text Books:

1. M.N.O.Sadiku,"ElementsofElectromagnetics",OxfordUniversityPublication,2014.
2. A.Pramanik,"Electromagnetism-Theoryandapplications",PHILearningPvt.Ltd,NewDelhi,2009.

Reference Books:

1. G.W.Carter,"The electromagnetic field in its engineering aspects",Longmans,1954.
2. W.J.Duffin,"ElectricityandMagnetism",McGrawHillPublication,1980.
3. W.J.Duffin,"AdvancedElectricityandMagnetism",McGrawHill,1968.
4. A.Pramanik, "Electromagnetism-Problems withsolution",PrenticeHallIndia,2012.

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EC2701

ANALOG ELECTRONIC CIRCUITS LAB

L-T-P-C

Externals: 60Marks
Internals: 40Marks **0-0-2-1**

Course Objectives:

- To provide practical Exposure for the students of semiconductor devices.
- Operation amplifiers and their application.
- Understanding of Rectifications.
- Understanding of Non linear wave shape.

Course Outcomes:

At the end of this course, students will be able to

- Understand the characteristics of transistors
- Design and analysis various rectifier and amplifier circuits.
- Design sinusoidal and non-sinusoidal oscillators.
- Design OP-AMP based circuits.

LIST OF EXPERIMENTS: (Any TEN of the following)

1. Study of Diode Characteristics.
2. Design of Clipping circuits.
3. Design of Clamping Circuits.
4. Design of Half rectifier and Full wave rectifier with and without filters.
5. Study of BJT Characteristics.
6. Design of CE and CC amplifiers and draw its frequency response Curve.
7. Study of FET Characteristics.
8. Design of Common source and Common Drain FET amplifiers and draw its frequency response Curve.
9. Design of Adder, Subtractor, Comparator using IC 741 Op-Amp.
10. Design of Integrator and Differentiator Using IC 741 Op-Amp.
11. Design of Wein - Bridge Oscillator using OP-AMP.
12. Design of RC Phase Shift Oscillator using OP-AMP.
13. Design of Monostable Multivibrator using IC555 Timer.
14. To generate triangular &square wave using operational Amplifier.

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ME2105

Engineering Mechanics

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 1 - 0 - 4**

Course Outcomes:

- To understand representation of force, moments for drawing free-body diagrams and analyze friction-based systems in static condition
- To locate the centroid of an area and calculate the moment of inertia of a section.
- Apply of conservation of momentum & energy principle for particle dynamics and rigid body kinetics
- Understand and apply the concept of virtual work, rigid body dynamics and systems under vibration

Unit I: Introduction to Engineering Mechanics

Force Systems Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Vector Mechanics- dot product, cross product, Problems

Unit II: Structural analysis and friction

Equilibrium in three dimensions; Method of Sections; Method of Joints; How to determine if a member is in tension or compression; Simple Trusses; Zero force members; Beams & types of beams; Frames & Machines, Problems
Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction, screw jack & differential screw jack, Problems

Unit III: Properties of surfaces and solids

Distributed Force: Centroid and Centre of Gravity; Centroids of a triangle, circular sector, quadrilateral, etc., Centroid of simple figures from first principle, centroid of composite sections; Centre of Gravity and its implications, Problems
Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections;

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Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook, Problems.

Unit IV: Virtual Work and Energy Method:

Virtual displacements, principle of virtual work for particle and ideal system of rigid bodies, degrees of freedom. Active force diagram, systems with friction, mechanical efficiency. Conservative forces and potential energy (elastic and gravitational), energy equation for equilibrium. Applications of energy method for equilibrium. Stability of equilibrium, Problems.

Unit V: Dynamics

Rectilinear motion; Plane curvilinear motion (rectangular, path, and polar coordinates). 3-D curvilinear motion; Relative and constrained motion; Newton's 2nd law (rectangular, path, and polar coordinates). Work-kinetic energy, power, potential energy. Impulse-momentum (linear, angular); Impact (Direct and oblique), Problems

Basic terms, general principles in dynamics; Types of motion, Instantaneous centre of rotation in plane motion and simple problems; D'Alembert's principle and its applications in plane motion and connected bodies.

Text books:

1. Irving H. Shames (2006), Engineering Mechanics, 4th Edition, Prentice Hall
2. F. P. Beer and E. R. Johnston (2011), Vector Mechanics for Engineers, Vol I - Statics, Vol II, – Dynamics, 9th Ed, Tata McGraw Hill.
3. R.C. Hibbler (2006), Engineering Mechanics: Principles of Statics and Dynamics, Pearson Press.
4. Andy Ruina and Rudra Pratap (2011), Introduction to Statics and Dynamics, Oxford University Press.
5. Shanes and Rao (2006), Engineering Mechanics, Pearson Education, 6. Hibler and Gupta (2010), Engineering Mechanics (Statics, Dynamics) by Pearson Education.

Reference books:

1. Reddy Vijaykumar K. and K. Suresh Kumar (2010), Singer's Engineering Mechanics
2. Bansal R.K. (2010), A Text Book of Engineering Mechanics, Laxmi Publications
3. Khurmi R.S. (2010), Engineering Mechanics, S. Chand & Co.
4. Tayal A.K. (2010), Engineering Mechanics, Umesh Publications

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EE2101

Electrical Machines-I

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 1 - 0 - 4**

Course Objectives:
This course introduces the concept of

- Construction operational features of energy conversion devices i.e., DC machines and transformers.
- Characteristics of DC machines and transformers and their applications

Course Outcomes: At the end of this course, students will demonstrate the ability to

- Understand the operation of dc machines.
- Analyze the differences in operation of different dc machine configurations.
- Analyze single phase and three phase transformers circuits

UNIT I: Magnetic fields, magnetic circuits and Magnetic Forces (10 Hours)

Review of magnetic circuits - MMF, flux, reluctance, inductance; review of Ampere Law and Biot Savart Law; B-H curve of magnetic materials, hysteresis and eddy current losses, Concept of statically and dynamically induced emf, Lorentz's Equation of Force Energy stored in the magnetic circuits; Field energy and mechanical energy, determination of mechanical force; flow of energy in electromechanical devices.

UNIT II: DC machines (8 Hours)

Basic construction of a DC machine, magnetic structure - stator yoke, stator poles, pole-faces or shoes, air gap and armature core, visualization of magnetic field produced by the field winding excitation with armature winding open, air gap flux density distribution, flux per pole, induced EMF in an armature coil. Armature winding and commutation - Elementary armature coil and commutator, lap and wave windings, construction of commutator, linear commutation Derivation of back EMF equation, armature MMF wave, derivation of torque equation, armature reaction, air gap flux density distribution with armature reaction.

UNIT III: DC machine - motoring and generation (8 Hours)

Armature circuit equation for motoring and generation, Types of field excitations – separately excited, shunt, series and compound. Open circuit characteristic of separately excited DC generator, back EMF with armature reaction, voltage build-up in a shunt generator, critical field resistance and critical speed, Characteristics of generators and motors: separately excited, shunt, series and compound. Speed control of DC motors, Losses, load testing and back-to-back testing of DC machines, Starting of DC motor

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UNIT IV: Transformers-1 (8 Hours)

Principle, construction and operation of single-phase transformers, equivalent circuit, phasor diagram, voltage regulation, losses and efficiency Testing - open circuit and short circuit tests, polarity test, back-to-back test, separation of hysteresis and eddy current losses, Autotransformers - construction, principle, applications and comparison with two winding transformer, Magnetizing current, effect of nonlinear B-H curve of magnetic core material, harmonics in magnetization current

UNIT V: Transformers-2 (8 Hours)

Three-phase transformer - construction, types of connection and their comparative features, Parallel operation of single-phase and three-phase transformers, Phase conversion – Scott connection, three-phase to six-phase conversion, Tap-changing transformers - No-load and on-load tap-changing of transformers, Three-winding transformers. Cooling of transformers.

Text books:

1. P. S. Bimbhra, " Electrical Machinery", Khanna Publishers, 2011.

2. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill Education, 2010.

Reference Books:

1. A. E. Fitzgerald and C. Kingsley, "Electric Machinery", New York, McGraw Hill Education, 2013.

2. A. E. Clayton and N. N. Hancock, "Performance and design of DC machines", CBS Publishers, 2004.

3. M. G. Say, "Performance and design of AC machines", CBS Publishers, 2002.

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EE2701 **ELECTRICAL MACHINES-1 LAB**

Externals:60Marks

L-T-P-C

Internals:40Marks **0-0-3-1.5**

Course Objectives:

- To expose the students to the operation of DC Generator
- To expose the students to the operation of DC Motor
- To examine the self excitation in DC generators
- To expose the students to the operation of transformers

Course Outcomes: After completion of this lab the student is able to

- Start and control the Different DC Machines
- Assess the performance of different machines using different testing methods
- Identify different conditions required to be satisfied for self - excitation of DC Generators.
- Separate iron losses of transformers

List of Experiments :

1. To obtain magnetization characteristics of a d.c. shunt generator.
2. Polarity and ratio test of single phase transformers.
3. Toobtainloadcharacteristicsofad.c.shuntgeneratorandcompoundgenerator
4. To obtain efficiency of a dc shunt machine using Swinburn's test.
5. To perform Hopkinson's test and determine losses and efficiency of DC machine.
6. To obtain speed-torque characteristics of a dc shunt motor.
7. To obtain speed control of dc shunt motor using
   (a) armature resistance control (b) field control
8. To obtain equivalent circuit, efficiency and voltage regulation of a single phase transformer using O.C. and S.C. tests.
9. To obtain efficiency and voltage regulation of a single phase transformer by Sumpner's test.
10. Load test on dc series generator
11. Field's test on dc series machine
12. Scott Connection of transformers

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CS2301 **DATA STRUCTURES**

Contact Hours per Week 40 **L- T -P -C**

Internal External Credits 60 **3- 0- 0- 3**

Course Objectives

- To develop proficiency in the specification, representation, and implementation of abstract data types and data structures.
- To get a good understanding of applications of data structures.
- To solve advanced computer science problems by making appropriate choice for intended applications.

Course Outcomes

- Students will be able to represent and manipulate linear and non-linear data structures using the C programming language.
- Students will be able to apply different searching and sorting algorithms to solve problems.
- Students will understand the concepts of binary trees and search trees, and be able to perform operations on them.
- Students will be able to implement stack and queue ADTs using both array and linked representations.
- Students will understand the basics of graph theory, including graph traversals such as BFS and DFS.

UNIT-I: Basic Concepts

Algorithm specification, Introduction, Recursive algorithms. Introduction to Linear and non-linear Data structures. Representation of single and two dimensional arrays, Singly Linked List Operations- Insertion, Deletion, Concatenating Single Linked Lists, Circular Linked List, Doubly Linked list.

UNIT-II: Stack ADT

Definitions, Operations, array and linked representation in C, application- infix to postfix conversion, postfix expression evaluation, recursion implementation.

Queue ADT- definitions and operations, circular queues, double ended queue array and linked representation.

UNIT-III: Trees

Terminology, Representation of Trees, Binary tree ADT, Properties of Binary trees, array and linked representation of Binary trees, Max Heap, Min Heap.

Graph- Introduction, Definition and terminology, Graph traversals- BFS and DFS.

UNIT-IV: Searching and sorting

Linear and Binary Search, Sorting – Insertion, Bubble, Selection, Radix, Quick, Merge, Heap sorts.

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Comparisons of Sorting Algorithms.

Hashing: Hash Table Representation, hash functions, collision resolution-separate chaining, open addressing-linear probing, quadratic probing, double hashing, rehashing, extendible hashing.

UNIT-V Search Trees

Binary search Trees-operations, AVL Trees-height of AVL, operations. Tree operations on B Trees and B+ trees. Red Black trees-Definition and Representation and application .

Text Books

1. N. Karumanchi, "Data Structures and Algorithms Made Easy," CareerMonk Publications, 2011.

References

1. Horowitz, E., Sahni, S., & Freed, S. A. (2007). Fundamentals of Data Structures in C, 2nd
2. Edition. Universities Press.
3. Tanenbaum, A. S., Langsam, Y., & Augenstein, M. J. (2011). Data Structures Using C. Pearson
4. Education India.
5. M. A. Weiss, "Data Structures and Algorithm Analysis in C++," 3rd ed., Pearson Education,
6. 2006.
7. M. T. Goodrich, R. Tamassia, D. Mount, "Data Structures and Algorithms in C++," 2nd ed.,
8. Wiley India Pvt. Ltd, 2004.
9. Department

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SECOND YEAR (E2) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE2201	Electrical Machines-II	PCC	4	1	0	4	4
2	EE2202	Power Electronics	PCC	3	1	0	4	4
3	EC2205	Digital Electronics	ESC	3	0	0	3	3
4	EE2203	Power Systems-I	PCC	3	0	0	3	3
5	EC2206	Signals and Systems	ESC	2	1	0	3	3
6	EE2801	Electrical Machines-II Lab	PCC	0	0	3	3	1.5
7	EE2802	Power Electronics Lab	PCC	0	0	3	3	1.5
8	EC2803	Digital Electronics Lab	ESC	0	0	2	2	1
9	HS2201	Essence of Indian Traditional Knowledge	MC	0	0	0	0	0
Total				15	3	10	27	21

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EE2201 **ELECTRICAL MACHINES-II**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-1-0-4**

Course Objectives:

- To deal with the detailed analysis of poly-phase induction motors and Alternators
- To understand operation, construction and types of single phase motors and their applications in house hold appliances and control systems.
- To introduce the concept of parallel operation of alternators
- To introduce the concept of regulation and its calculations.

Course Outcomes: At the end of this course, students will demonstrate the ability to

- Understand the concepts of rotating magnetic fields.
- Understand the operation of ac machines.
- Analyze performance characteristics of ac machines.
- Understand the concept of speed control in induction motors.

UNIT I: **Fundamentals of AC machine windings** (10 **Hours)**

Physical arrangement of windings in stator and cylindrical rotor; slots for windings; concentrated winding, distributed winding, single layer winding, full-pitch coils, pitch factor, distribution factor, elimination of harmonics, Air-gap MMF distribution with fixed current through winding - concentrated and distributed.

UNITII: Pulsating and revolving magnetic fields (6 Hours)

Constantmagneticfield,pulsatingmagneticfield-alternatingcurrentinwindingswithspatial displacement, Magnetic field produced by a single winding - fixed current and alternating current, Pulsatingfieldsproducedbyspatiallydisplacedwindings,Windingsspatiallyshiftedby 90 degrees, Addition of pulsating magnetic fields, Three windings spatially shifted by 120 degrees (carrying three-phase balanced currents), revolving magnetic field.

UNIT III: Three Phase Induction Machines (18 Hours)

Construction, Types (squirrel cage and slip-ring), Equivalent circuit, Phasor Diagram, Torque Slip Characteristics, Starting and Maximum Torque, Losses and Efficiency. Effect of parameter variation on torque speed characteristics (variation of rotor and stator resistances, stator voltage, frequency). Methods of starting, braking and speed control for induction motors, Concept of Cogging and Crawling, Double cage rotor induction motor, Testing of induction motor, Circle diagrams. Generator operation, Self-excitation, Doubly-Fed Induction Machines.

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UNITIV: Single-phase induction motors (6 Hours)

Constructional features double revolving field theory, equivalent circuit, and determination of parameters. Split-phase starting methods and applications

UNIT V: Synchronous machines (15Hours)

Constructional features, cylindrical rotor synchronous machine , generated EMF, equivalent circuit and phasor diagram, armature reaction, synchronous impedance, voltage regulation, Operating characteristics of synchronous machines, V & inverted V curves. Salient pole machine - two reaction theory, analysis of phasor diagram, power angle characteristics, Parallel operation of alternators - synchronization and load division, synchronous motor & it's starting methods.

Text Books:

1. P. S. Bimbhra, " Electrical Machinery", Khanna Publishers,2011.
2. I.J.NagrathandD.P.Kothari,"ElectricMachines",McGrawHillEducation,2010.

Reference Books:

1. S.Langsdorf,"Alternatingcurrentmachines",McGrawHillEducation,1984.
2. P.C.Sen,"PrinciplesofElectricMachinesandPowerElectronics",JohnWiley&Sons, 2007

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EE2801

ELECTRICAL MACHINES-II LAB

Externals:60Marks **L-T-P-C**

Internals:40Marks **0-0-3-1.5**

Course Objectives:

- To understand the operation of synchronous machines
- To understand the analysis of power angle curve of a synchronous machine
- To understand the equivalent circuit of a single phase transformer and single phase induction motor
- To understand the circle diagram of an induction motor by conducting a blocked rotor test.

Course Outcomes: After completion of this lab the student is able to

- Assess the performance of different machines using different testing methods
- To convert the Phase from three phase to two phase and vice versa
- Compensate the changes in terminal voltages of synchronous generator after estimating the change by different methods
- Control the active and reactive power flows in synchronous machines
- Start different machines and control the speed and power factor

List of Experiments :

1. To perform no load and blocked rotor tests on a three phase squirrel cage induction motor and determine equivalent circuit.
2. To perform load test on a three phase induction motor and draw Torque- peed characteristics.
3. To determine speed-torque characteristics of three phase slip ring induction motor and study the effect of including resistance in the rotor circuit.
4. Todeterminespeed-torquecharacteristicsofsinglephaseinductionmotorandstudythe effect of voltage variation.

Electrical and Electronics Engineering Dept

5. To perform no load and blocked rotor tests on a single phase induction motor and determine equivalent circuit.
6. Determine voltage regulation at full load and at unity, 0.8 lagging and leading power factors by
   a. EMFmethod
   b. MMFmethod.
7. To study synchronization of an alternator with the infinite bus by using:
   a. dark lamp method
   b. two bright and one dark lamp method.
8. To determine V-curves and inverted V-curves of a three phase synchronous motor.
9. Determine Xd and Xq of a three phase salient pole synchronous machine using the slip test and to draw the power-angle curve.

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EE2202 **POWER ELECTRONICS**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-1-0-4**

Course Objectives: This course will develop students' knowledge in/on

- Characteristics and applications of basic power semi conductor switches
- Controlled rectifier circuits, DC-DC converter, inverter, AC voltage controller and cyclo converters
- To familiarize students to the principle of operation, design and synthesis of different power conversion circuits and their applications.
- To provide strong foundation for further study of power electronic circuits and systems.

Course Outcomes: At the end of this course students will demonstrate the ability to

- Understand the differences between signal level and power level devices.
- Analyze controlled rectifier circuits.
- Analyze the operation of DC-DC choppers.
- Analyze the operation of voltage source inverters

UNIT-I: Power switching devices(8 Hours)

Diode, Thyristor, BJT, MOSFET,IGBT: I-V Characteristics and switching characteristics; Firing circuit for thyristor; Voltage and current commutation of a thyristor; Gate drive circuits for MOSFET and IGBT, Daic and Traic.

UNIT-II: Thyristor rectifiers(12Hours)

Single-phase half-wave and full-wave rectifiers, Single-phase full-bridge thyristor rectifier with R, RL and RLE loads; Three-phase full-bridge thyristor rectifier with R, RL loads; Input current wave shape and power factor, effect of source impedance and dual converter.

UNIT-III:DC-DC converter(8Hours)

Elementary chopper with an active switch and diode, concepts of duty ratio and average voltage, power circuit of abuckconverter,boostconverterandbuck-boostconverteranalysisandwaveforms at steady state, relation between duty ratio and average output voltage, Voltage ripple and current ripple, introduction to isolated DC-DC converters.

UNIT-IV: Inverter (10 Hours)

Single-phase voltage source inverter, three-phase voltage source inverter (180 & 120 degree conduction modes), modulation techniques(PWM,SPWM),current source inverter

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UNIT-V: AC voltage controller and cyclo converters (5 hours)

Principle of phase control, principle of integral cycle control, single phase voltage controllers; principle of cyclo-converter operation, single phase to single phase cyclo-converter and single phase to three phase cyclo-converter.
Applications: Battery Charger, UPS and SMPS

TEXT BOOK:

1. Dr. P.S Bimbhra, "Power Electronics", Khanna Publishers,2012
2. M.H. Rashid, "Power Electronic Devices, Circuits and Application" Pearson Education India,2009.

Reference Books:

1. R.W. Erickson and D. Maksimovic, "Fundamentals of Power electronics", Springer Science &Business Media,2007.
2. L. Umanand, "Power Electronics: Essentials and Application", Wileey India,2009.

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EE2802

POWER ELECTRONICS LAB

Externals:60Marks **L-T-P-C**

Internals:40Marks **0-0-3-1.5**

Course Objectives: To Provide Practical exposure on

- Characteristics of basic power semiconductor switches
- Applications of basic power semiconductor switches like controlled rectifier circuits,
- DC-DC converter, inverter, AC voltage controller etc.
- Examine the characteristics of various devices and application of firing circuits used in power electronics

Course Outcomes: After completion of this laboratory course, students will be able to

- Determine the power semiconductor switches characteristics and their applications
- Design gate firing & commutation circuits for SCRs.
- Analyze the operation of converters, inverters and choppers.
- Design and simulate power electronic circuits and plot their characteristics.

LISTOFEXPERIMENTS:

1. Study of Characteristics of SCR, MOSFET & IGBT
2. Gate firing circuits for SCR's
3. Single Phase Semi-converter with Rand RL load
4. Single Phase fully controlled bridge converter with R and RL loads
5. Single Phase dual converter with RL loads
6. Three Phase Semi-converter with R-load
7. Three Phase Bridge converter with R and RL loads
8. Isolated DC-DC converter
9. Single phase half bridge and full bridge inverter
10. Buck Converter with R and RL loads
11. Boost Converter with R and RL loads
12. Single Phase AC Voltage Controller with R and RL Loads
13. Single Phase Cyclo converter with R and RL loads

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EC2205

DIGITAL ELECTRONICS

Externals: 60Marks **L-T-P-C**
Internals: 40Marks **3-0-0-3**

Course Objectives: To provide practical exposure on

- Various combinational and sequential circuits and filters.
- Applications of Operational Amplifier as adder, integrator
- Applications of Operational Amplifier as voltage to current converters.
- To understand the concept of Sequential circuits.

Course Outcomes:

Upon completion of this course the student will be able to

- Design counters, NAND gate and adders.
- Design multiplexer, 7-segment LED display.
- Analyze the application of Operational Amplifier as adder, integrator and voltage to current converters.
- Design of Low pass, High pass and Band pass Filters

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UNIT I: Circuits and logic families (10 Hours)

Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, examples of IC gates, number systems-binary, signed binary, octal hexadecimal number, binary arithmetic, one's and two's complements arithmetic, codes, error detecting and correcting codes, characteristics of digital ICs, digital logic families, TTL, Schottky TTL and CMOS logic, interfacing CMOS and TTL, Tri-state logic.

UNIT II : Combinational Digital Circuits (10 Hours)

Standard representation for logic functions, K-map representation and simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractions, weighted and Non weighted codes, BCD, Gray codes, carry look ahead adder, serial adder, digital comparator, parity checker/generator, code converters, priority encoders, decoders.

UNIT III: Sequential circuits and systems (10 Hours)

Bit memory, the circuit properties of Bistable latch, the clocked SR flip flop, J- K, T and D types flip-flops, applications of flip-flops, shift registers, applications of shift registers, serial to parallel converter, parallel to serial converter, ring counter, sequence generator, ripple(Asynchronous) counters, synchronous counters, counters design using flip flops , asynchronous sequential counters, applications of counter.

UNIT IV: A/D and D/A Converters (7Hours)

Analog to Digital converters: Quantization and encoding, parallel comparator A/D converter, successive approximation A/D converter, counting A/D converter, dual slope A/D converter, A/D converter using voltage to frequency and voltage to time conversion, specifications of A/D converters, example of A/D converter ICs.
Digital to analog converters: weighted resistor/converter, R-2R Ladder D/A converter, specifications for D/A converters, examples of D/A converter ICs, sample and hold circuit, analog to digital converters.

UNIT V: Semiconductor memories and Programmable logic devices. (5 Hours)

Memory organization and operation, expanding memory size, classification and characteristics of memories, sequential memory, read only memory (ROM), read and write memory (RAM)

TEXT BOOKS:

1. Robert L Boylested and Louis Nashelsky, Electronic Devices and Circuit Theory, 10th ed., New Delhi: Pearson Ind. Pvt. Ltd., 2009
2. Jacob Millman, Christos C Halkias & Satyabrata JIT, Electronic Devices and Circuits, New Delhi: Tata McGraw Hill Education (INDIA) Private Ltd, 2007.

REFERENCE BOOKS:

1. Electronic Devices Conventional and current version -Thomas L. Floyd 2015, pearson.
2. Linear Integrated Circuits 4th Edition By Dr. Roy Choudhury, shail B.Jain.
3. P.R. Gray, R.G Meyer and S. Lewis, “Analysis and Design of Analog Integrated Circuits”, John Wiley &Sons,2001

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EE2203 **POWER SYSTEMS-I**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives: This course introduce

- To understand the power generation through conventional and non- conventional sources.
- To know about the substations, Overhead Line Insulators and underground cables.
- To know about the corona, sag and AC & DC distribution systems.
- To illustrate the economics aspects of power generation, tariff methods.

Course Outcomes: After completion of this course, students will be able to

- Describe the operation of conventional generating stations
- Describe about the different types of substations available
- Determine Different Types of Tariff's in power system
- Design Distribution of voltage along the string insulators and design concept of underground cables & Solve Problems.

UNIT-I: Conventional and Non-conventional Energy Sources (8 Hours)

Introduction: Typical Layout of an Electrical Power System, Present Power Scenario in India. Conventional Sources (Qualitative): Hydro station, Steam Power Plant, Nuclear Power Plant and Gas Turbine Plant.

Non Conventional Sources (Qualitative): Ocean Energy, Tidal Energy, Wave Energy, wind Energy, Fuel Cells, and Solar Energy.

UNIT-II: Outdoor and Indoor Substations (6 Hours)

Air insulated substations - Indoor, Outdoor, layout, substation equipment. Bus bar arrangements in the Sub-Stations: Single bus bar, sectionalized single bus bar, main and transfer bus bar system with relevant diagrams.

Gas insulated substations (GIS) - Advantages, various types, single line diagram, bus bar, construction aspects, Installation and maintenance of GIS, Comparison of Air insulated substations and Gas insulated substations.

UNIT-III: Overhead Line Insulators and Insulated Cables (10 Hours)

Overhead Line Insulators: Introduction, types of insulators, Potential distribution over a string of suspension insulators, string efficiency, methods of equalizing the potential, testing of insulators.

Insulated Cables: Introduction, insulation, insulating materials, Extra high voltage cables, grading of cables, insulation resistance of a cable, Capacitance of a single core and three core cables, Overhead lines versus underground cables, types of cables.

UNIT-IV: Corona and Sag (10 Hours)

Corona: Introduction, disruptive critical voltage, corona loss, Factors affecting corona loss and methods of reducing corona loss, Disadvantages of corona, interference between power and Communication lines.

Sag: The Catenary curve, Sag Tension calculations, Supports at Different Levels, Stringing chart, Sag

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template, Equivalent span, Stringing of conductors, Vibration and Vibration dampers.

UNIT-V: Economics of Generation and Distribution (8 Hours)

Economics of Generation: Introduction, definitions of connected load, maximum demand, demand factor, load factor, diversity factor, plant utilization factor, plant capacity factor, Load duration curve, number and size of generator units. Base load and peak load plants. Cost of electrical energy- calculation of energy, fixed cost, running cost, Tariff on charge to customer.

A.C. Distribution: Introduction, Single phase, 3-phase, 3 phase 4 wire system, bus bar arrangement, Selection of site for substation.

D.C. Distribution: Calculations, uniformly loaded distributor fed at one end, distributor fed at both ends, distributor with both concentrated and uniform loading, ring distributor and with Interconnector.

Text books:

1. C.L. Wadhwa –"Generation, Distribution and Utilization of Electrical Energy", Second Edition, New Age International, 2009
2. C.L. Wadhwa –"Electrical Power Systems", Fifth Edition, New Age International, 2009 Reference

References:

1. W.D.Stevenson –"Elements of Power System Analysis", Fourth Edition, McGraw Hill, 1984.
2. M.V. Deshpande –"Elements of Electrical Power Station Design", Third Edition, Wheeler Pub. 1998
3. H.Cotton & H. Barber-"The Transmission and Distribution of Electrical Energy", Third Edition, ELBS, B.I.Pub., 1985
4. Syed A Nasar, "Electric Power Systems", Mcgraw-Hill, 1/e, 2006

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EE2206

SIGNALS AND SYSTEMS

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **2-1-0-3**

Course Objectives: This course introduces

- Concepts of signals and systems and their characteristics
- Various mathematical tools like Fourier, Laplace and z- transforms to analyze an LTI systems

Course Outcomes: At the end of this course, students will demonstrate the ability to

- Understand the concepts of continuous time and discrete time systems.
- Analyze systems in complex frequency domain.
- Understand sampling theorem and its implications.

UNIT I: Introduction to Signals and Systems (6 hours):

Signals and systems as seen in everyday life, and in various branches of engineering and science. Continuous and discrete time signals, continuous and discrete amplitude signals, properties of signal, Power and energy of a signal, some special signals of importance: unit step, unit impulse, ramp, parabolic sinusoid, complex exponential,. System properties: linearity: additivity and homogeneity, shift-invariance, causality, stability, reliability. Examples.

UNIT II: Behavior of continuous and discrete-time LTI systems (8 hours)

Impulse response and step response, convolution, input-output behavior with a periodic convergent input, cascade interconnections. Characterization of causality and stability of LTI systems. System representation through differential equations and difference equations. Periodic inputs to an LTI system, the notion of a frequency response and its relation to the impulse response.

UNIT III: Fourier Transforms(8 hours)

Fourier series representation of periodic signals, Waveform Symmetries, Calculation of Fourier Coefficients. Properties of Fourier series, Fourier Transform, convolution/multiplication and their effect in the frequency domain, magnitude and phase response, Properties of Fourier Transforms, Fourier domain duality. The Discrete- Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT). Parseval's Theorem.

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UNIT IV: Laplace and z-Transforms (10 Hrs)

Review of the Laplace Transform for continuous time signals and systems, system functions, poles and zeros of system functions and signals, Laplace domain analysis, solution to differential equations and system behavior. The z-Transform for discrete time signals and systems, system functions, poles and zeros of systems and sequences, z-domain analysis.

UNIT V: Sampling and Reconstruction (6 hours)

The Sampling Theorem and its implications. Spectra of sampled signals. Reconstruction: ideal interpolator, zero-order hold, first-order hold. Aliasing and its effects. Relation between continuous and discrete time systems. Introduction to the applications of signal and system theory: modulation for communication, filtering, feedback control systems.

Text Books:

1. A. V. Oppenheim, A. S. Willsky and S. H. Nawab, "Signals and systems",Prentice Hall India, 1997.
2. B. P. Lathi, " Linear Systems and Signals" , Oxford University Press, 2009.
3. M. J. Robert " Fundamentals of Signals and Systems", McGraw HillEducation, 2007.

Reference Books:

1. J. G. Proakis and D. G. Manolakis, " Digital SignalProcessing: Principles, Algorithms, and Applications", Pearson, 2006.
2. H. P. Hsu, " Signals and systems" , Schaum's series, McGraw Hill Education, 2010.
3. S. Haykin and B. V. Veen, " Signals and Systems" , John Wiley and Sons, 2007.
4. A. V. Oppenheim and R. W. Schafer, " Discrete-Time Signal Processing" , Prentice Hall, 2009.

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EC2803

DIGITAL ELECTRONICS LAB

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **0-0-2-1**

Course Objectives:

To provide practical exposure on

- Various combinational and sequential circuits and filters.
- Applications of Operational Amplifier as adder.
- Applications of integrator and voltage to current converters.
- Understanding of and design of filters working

Course Outcomes:

Upon completion of this course the student will be able to

- Design counters, NAND gate and adders.
- Design multiplexer, 7-segment LED display and LPF, HPF, BPF
- Analyze the application of Operational Amplifier as adder.
- Integrator and voltage to current converters.

LIST OF EXPERIMENTS: Any TEN of the following experiments

1. Design of a counter asynchronous and synchronous
2. I/O characteristics of a NAND gate
3. Design of a full adder circuit
4. Design of a digital comparator
5. Simplification Boolean function using K-map
6. Design of a multiplexer
7. Design of a 7-segment LED display To study application of Operational Amplifier as adder, integrator and voltage to current converters.
8. Design of filters
9. To design a low pass filter Second order filters using operational amplifier for cutoff frequency 1 KHz.
   i. To design a high pass filter Second order filters using operational amplifier for frequency 12 KHz.
   ii. To design a band pass filter with unit gain of pass band from 1 KHz to 12 KHz.
10. To study application of Operational Amplifier as voltage comparator.
11. To generate triangular & square wave using operational amplifier.
12. To study regulation of unregulated power supply using IC 7805/7812 voltage regulator and measure the load and line regulations

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HS2201 ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE

Internals: 40 Marks **L-T-P-C**
Externals: 60 Marks **2-0-2-0**

UNIT –I: BASIC STRUCTURE OF INDIAN KNOWLEDGE SYSTEM

VEDA (AYURVEDA, DHANURVEDA, GANDHARVA VEDA, STHAPATYA AATI **(SHILPA VEDA**), ARTHA VEDA, VEEDANGA (SHIKSHA, KALPA, CHHANDA, NIRUKTHA, VYAKARANA, JYOTHISHYA) DARMA SHASTRA, MIMASHA, PURANA, TARKASHASTRA

UNIT – II: MODERN SCIENCE AND INDIAN KNOWLEDGE SYSTEM

YOGA HOLISTIC HEALTH CARE

UNIT – III: INDIAN PHILOSOPHICAL TRADITION:

A) ORTHODOX (HINDU) SCHOOL: SAMKYA, YOGA, NYAYA, VAISHESHIKA, PURVA MIMAMSA, VEDHANTA,
B) HETORODOX (NON-HINDU) SCHOOLS: CARVAKA, JAIN, BUDDHA

UNIT-IV: INDIAN LINGUISTIC TRADITION

PHONOLOGY, MORPHOLOGY, SYNTAX AND SEMANTICS

UNIT –V: INDIAN ARTISTIC TRADITION

CHITRA KALA, MANTRA KALA, VAASTU KALA, SANGEETHA KALA, NRUTHYU EVAM SAHITYAM

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THIRD YEAR (E3) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE3101	Power Systems-II	PCC	3	1	0	4	4
2	EE3102	Control Systems	PCC	3	1	0	4	4
3	EE3103	Electrical Measurements and Instrumentation	PCC	3	1	0	4	4
4	EC3104	Micro Processors & Micro Controllers	ESC	3	0	0	3	3
5	EE3701	Control Systems Lab	PCC	0	0	2	2	1
6	EE3702	Electrical Measurements and Instrumentation Lab	PCC	0	0	2	2	1
7	EC3703	Micro Processors & Micro Controllers Lab	ESC	0	0	2	2	1
8	HS3101	Corporate Communication	HSMC	0	0	2	2	1
Total				12	3	6	21	19

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EE3101 **Power Systems-II**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-1-0-4**

Course Objectives: To provide the knowledge to

- Analyze transmission line performance
- Apply load compensation techniques to control reactive power
- Understand the application of per unit quantities
- Determine the fault currents for symmetrical and unbalanced faults

Course Outcomes: After completion of this course, students will be able to

- Analyze circuit parameters of transmission lines & transmission line performance & Solve Problems.
- Describe the voltage control methods and different compensation methods available
- Explain the significance of per unit quantities.
- Determine the fault currents for symmetrical and unbalanced faults

UNIT I: Transmission Line Parameters (10 Hours)

Inductance and Capacitance Calculations of Transmission Lines: Line conductors, inductance and capacitance of single phase and three phase lines with symmetrical and unsymmetrical spacing, Composite conductors-transposition, bundled conductors, and effect of earth on capacitance, Skin and Proximity effect.

UNIT II: Performance of Transmission Lines (8 Hours)

Representation of lines, short transmission lines, medium length lines, nominal T and π representations, long transmission lines. The equivalent circuit representation of a long Line, A, B, C, D constants, Ferranti Effect, Power flow through a transmission line, receiving end power circle diagram.

UNIT III: Voltage Control, Power factor Improvement & Compensation in Power Systems (8 Hours)

Voltage Control: Introduction – methods of voltage control, shunt and series capacitors / Inductors, tap changing transformers, synchronous phase modifiers, power factor improvement methods.
Introduction to compensation in power systems: Load ability characteristics of overhead lines, Uncompensated transmission line, Symmetrical line, Radial line with asynchronous load, Compensation of lines.

UNIT IV: Travelling Waves on Transmission Lines and Per Unit Representation (10 Hours)

Travelling Waves on Transmission Lines: Production of traveling waves, open circuited line, short circuited line, line terminated through a resistance, line connected to a cable, reflection and refraction at T-junction line terminated through a capacitance, capacitor connection at a T-junction, Attenuation of travelling waves.
Per Unit Representation of Power Systems: The single line diagram, impedance and reactance diagrams, per unit quantities, changing the base of per unit quantities, advantages of per unit system.

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UNIT V: Symmetrical Components and Fault Analysis (8 Hours)

Significance of positive, negative and zero sequence components, Average 3-phase power in terms of symmetrical components, sequence impedances and sequence networks, fault calculations, sequence network equations, single line to ground fault, line to line fault, double line to ground fault, three phase fault, faults on power systems, faults with fault impedance, reactors and their location, short circuit capacity of a bus.

Text Books:

1. C.L. Wadhwa: Electrical Power Systems – New Age International Pub. Co. Third Edition, 2001.
2. D.P. Kothari and I.J. Nagrath, Modern Power System Analysis - Tata McGraw Hill Pub. Co., New Delhi, Fourth edition, 2011 Reference

References:

1. John J. Grainger & W.D. Stevenson: Power System Analysis – McGraw Hill International 1994.
2. Hadi Saadat: Power System Analysis – Tata McGraw Hill Pub. Co. 2002
3. W.D. Stevenson: Elements of Power system Analysis – McGraw Hill International Student Edition.
4. Miller “Reactive power control in Electric systems “Wiley , 2/e ,2011

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EE3102

CONTROL SYSTEMS

Externals: 60Marks **L-T-P-C**
Internals: 40Marks **3-1-0-4**

Course Objectives:

- To introduce the principles and applications of control systems in everyday life
- The emphasis of this course is laid on stability analysis and design aspects of Control systems using classical control theory approaches.
- To understand the time response and frequency response of system
- To introduce the controllers like P, I, D, PI, PD, PID.

Course Outcomes: After completion of this course, students will be able to

- Determine the transfer function model and state variable models of electrical, mechanical systems
- Determine the stability of the LTI systems using time domain approaches like Routh-Hurwitz Criteria and Root locus.
- Determine the stability of the LTI systems using frequency domain approaches like bode plot, Nyquist plot.
- Determine controllability and observability of an LTI system described by a state variable model

UNIT-I: INTRODUCTION TO CONTROL SYSTEMS (10 Hours)

Introduction to control systems, Mathematical models of physical systems, Electrical Analogies of Mechanical Systems vice versa, Transfer function models of linear time-invariant systems, Feedback Control: Open-Loop and Closed-loop systems, Benefits of Feedback, Block diagram reduction technique, Signal Flow Graph.

UNIT II: TIME RESPONSE ANALYSIS (12 Hours)

Standard test signals, Time response of first and second order systems for standard test inputs, Application of initial and final value theorem, Design specifications for second-order systems based on the time-response, Concept of Stability, Routh-Hurwitz Criteria, Relative Stability analysis, Root-Locus technique, construction of Root-loci.

UNIT III: FREQUENCY-RESPONSE ANALYSIS (12 Hours)

Design specifications in frequency-domain, Relationship between time and frequency response, gain and phase margin, Bode plots, Polar plots, Nyquist stability criterion, Relative stability using Nyquist criterion, Closed-loop frequency response.

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UNIT IV: INTRODUCTION TO CONTROLLER DESIGN (8 Hours)

Steady-state accuracy, transient accuracy, disturbance rejection, insensitivity and robustness of control systems, Root-loci method of feedback controller design, Frequency domain methods of design. Proportional, Integral and Derivative Controllers, Lead and Lag compensation in designs, Effects of P, PD, Pl and PID controllers, Analog and Digital implementation of controllers.

UNIT V: STATE VARIABLE ANALYSIS AND APPLICATIONS OF CONTROL SYSTEMS (8 Hours)

Concepts of state variables, State space model, Diagonalization of State Matrix, Solution of state equations, Eigen values and Stability Analysis, state space representation of higher order differential equations, Concept of controllability and observability.
Applications: Synchros, magnetic amplifier, DC & AC servo motors, PLC etc

Text Books:

1. M.Gopal, "Control Systems: Principles And Design", McGraw Hill Education, 1997.
2. Norman S.Nice, "Control System Engineering", Norman S.Nice , 1995.

Reference Books:

1. Ogata, "Modern Control Engineering", Prentice Hall, 1991.
2. Nagrath and M. Gopal, "Control Systems Engineering", New Age International, 2009.

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EE3103 **ELECTRICAL MEASUREMENTS AND INSTRUMENTATION**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-1-0-4**

Course Objectives:

- To understand the operating principle of various types of analog instruments for measuring voltage, current, power, phase, frequency and energy.
- To determine the circuit parameters using AC and DC bridges.
- To understand operating principles of electronic measuring instruments

Course Outcomes: Upon competition of the course students will be able to

- Compare performance of MC, MI and Dynamometer types of measuring instruments, Energy meters and CRO
- Compute the errors in CTs and PTs.
- Selection of transducers for the measurement of temperature, displacement and strain.

UNIT I : Measurement of Voltage and Current (8 Hours)

Introduction: Methods of measurement, Measurement system, Classification of instrument systems, Characteristics of instruments & measurement systems, Definition of accuracy, precision, resolution. Speed of response. Errors in measurement& its analysis. Loading effect due to shunt and series connected instruments.

Measurement of Voltage and Current: General features, Construction, principle of operation and torque equation of Permanent magnet moving coil (PMMC), Moving Iron (MI), electrodynamometer, Induction, Thermoelectric and rectifier type instruments. Extension of instrument ranges using shunt, multipliers.

UNIT II: Measurement of Electrical Power and Energy (10 Hours)

Measurement of power: Construction and principle of operation of Electrodynamometer type wattmeter. Errors in Electrodynamometer type wattmeter. Low power factor wattmeter. Measurement of power in single phase system.

Measurement of phase, frequency and energy: Single phase and three phase electro dynamometer power factor meter, moving iron power factor meter. Construction and operation of different types of frequency meters. Construction and principle of operation of Single phase induction type energy meters, errors in energy meter and their compensation methods. Testing of energy meter by phantom loading method.

Instrument Transformers: CT and PT; their errors, Applications of CT and PT in the extension of instrument range.

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UNIT III: Measurement of Electrical Parameters (8 Hrs)

Measurement of resistance: Measurement of low resistance by Kelvin's double bridge, measurement of medium resistance by wheat stone bridge, voltmeter and ammeter method, substitution method and ohmmeter method. measurement of high resistance by loss of charge method, direct deflection method and Meggar.
Measurement of inductance and capacitance: Measurement of inductance with the help of AC Bridges (Maxwell's Inductance, Anderson, Hay's and owen's bridges)Measurement of capacitance with the help of AC Bridges (De Sauty's, Schering Bridge) their Applications and Limitations. Q meter.

UNIT IV: Potentiometers, Sensors & Transducers (8 Hours)

Principle of operation and application of Crompton's DC potentiometer, Polar and co-ordinate type of AC potentiometers. Magnetic Measurement- Ballistic galvanometer, Flux meter, Determination of hysteresis loop, measurement of iron losses.
Introduction to sensors & transducers, RTD, Thermistors, LVDT, Strain Gauge, Piezoelectric Transducers, Hall effect sensors. Flow measurement using magnetic flow measurement.

UNIT V: Digital Measurement of Electrical Quantities (8 Hours)

Advantages of digital instruments. Concept of digital measurement. Block diagram and theory of digital voltmeter, digital Frequency meter, Spectrum analyzer and harmonic distortion analyzers. Block diagram and working of Cathode Ray Oscilloscope, Cathode Ray Tube (CRT) & its components, Applications of CRO in measurement, Lissajous Pattern, Dual trace & dual beam oscilloscopes

Text Books:

1. A. K. Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Sons, India
2. J. B. Gupta, "Electrical Measurement & Measuring Instrument", S. K. Kataria & Sons

Reference Books:

1. E. W. Golding & F. C. Widdis, "Electrical Measurement &Measuring Instrument", A. W. Wheeler & Co. Pvt. Ltd. India
2. Forest K. Harris, "Electrical Measurement", Willey Eastern Pvt. Ltd. India

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EE3702 **ELECTRICAL MEASUREMENTS AND INSTRUMENTATION LAB**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **0-0-2-1**

Course Objectives:

This Laboratory course will develop student's knowledge in/on

- To provide students with good depth of knowledge of electrical and electronic measuring instruments.
- To understand measurement errors and non ideal electrical devices.

LIST OF EXPERIMENTS:

1. Measurement of low Resistance by using Kelvin's double bridge
2. Measurement of Inductance by Maxwell Bridge
3. Measurement of Inductance by Hay's Bridge
4. Measurement of Inductance by Anderson Bridge
5. Measurement of Capacitance by De sauty Bridge
6. Measurement of Capacitance by Schering Bridge
7. DC Crompton potentiometer
8. Dielectric Strength oil testing using H.T Testing kit
9. Calibration of LPF wattmeter by Phantom testing
10. Calibration of PMMC ammeter and PMMC voltmeter by using slide wire potentiometer
11. Calibration of single phase energy meter
12. Calibration of Dynamometer type power factor meter
13. C.T testing by comparison measurement of % ratio error and phase angle error
14. P.T testing by comparison measurement of % ratio error and phase angle error

COURSE OUTCOMES:

At the end of the course the student will be able to:

- Calibrate single phase energy meters
- Measure Resistance, Inductance and capacitance using AC and DC bridges
- Measure frequency, voltage peaks, phase difference with an oscilloscope.
- Compare performance of MC, MI and Dynamometer types of measuring instruments, Energy meters and CRO

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EC3104

MICROPROCESSORS AND MICROCONTROLLERS

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- To understand about 8085 microprocessor architecture, Instruction set and addressing modes.
- To know the use of inter facing devices and process of interfacing.
- To understand about 8051micro controller architecture.
- To know the microcontroller programs and interface devices

Course Outcomes: At the end of the course the students will be able to

1. Understand 8085microprocessor architecture and its operation.
2. Write assembly language program for a given task.
3. Interface memory and I/O devices to 8085 using peripheral devices.
4. Understand microcontrollers uses and their applications

UNIT-I:MicroprocessorArchitecture (10hrs)

Microprocessors, Microcomputers, and Assembly Language, Architecture Details and its operation, Bus organization of 8085, Registers, Memory unit of 8085, Instruction decoding & execution, 8085-Based single board Microcomputer, Pin out Diagram of 8085, Bus timings, 8085 Interrupts (Hardware and Software), 8085 Vectored Interrupts.

UNIT-II: 8085Programming (8hrs)

8085ProgrammingModel,OperandTypes,InstructionFormat,Addressing Modes, Instruction set, Writing and debugging, simple assembly Language Programs, Delays.

UNIT-III: Interfacing (10hrs)

Memory and I/O interfacing, Programmable Peripheral Interface 8255(PPI), Interfacing seven segment display, Interfacing matrix keyboard, A/D and D/A interfacing, Programmable Interval Timer (8253), Programmable Interrupt Controller (8259). Synchronous and Asynchronous Communication. RS232, SPI, I2C. Introduction to protocols like Blue-tooth and Zig-bee.

UNIT-IV: Microcontroller Architecture (10hrs)

Types of Microcontrollers, 8051 Microcontroller – Architecture, Memory organization, special function registers, pins and signals, timing and control, Ports and circuits, Counters and timers, Serial data input/output, Interrupts &timers, crystal oscillator.

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UNIT-V: 8051 Programming (10hrs)

The8051-programming model, Operand Types, Instruction cycle, addressing modes, 8051 instruction set, Classification of instructions. Simple programs and I/O interfacing.

Introduction to Advanced microcontrollers: Arduino programming and it's applications.

Text books:

1. Ramesh S. Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publishing,FifthEdition,2011.
2. Krishna Kant- Microprocessors and Microcontrollers- Architecture, Programming andSystemDesign8085,8086,8051,8096,Prentice-HallIndia-2007.

Reference books:

1. Kenneth.J.Ayala–The8051Microcontroller Architecture Programming and Applications", Thomson publishers, 2ndEdition,2007,
2. A.K. Ray & Bhurchandi, Advanced Microprocessors and Peripherals,TataMcGrawHill,2003

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HS3101 CORPORATE COMMUNICATION

Externals: 60 Marks L-T-P-C

Internals: 40 Marks 0-0-2-1

Pre-Requisites: None

Course Outcomes:

At the end of the course, the student will be able to:

- Understand corporate communication culture
- Prepare business reports and proposals expected of a corporate professional
- Use fluent and appropriate speech in formal business situations
- Develop good listening skills and leadership qualities
- Acquire appropriate corporate email, mobile and telephone etiquette
- Demonstrate corporate social responsibility and ethics

Chapter-1 Importance of Communication in the Corporate World:

What is corporate? - Corporate culture & communication - Process of communication – Networks & channels of communication – Barriers to communication – Strategies to overcome them - Use of technology in successful communication – Role of psychology in communication- Internal & External Communication.

Oral Communication

A. Oral Fluency and Communication Techniques Speech mechanics – Mental process of speaking – Extempore speech practice

–Body Language – Group discussion practice – Group dynamics – interview strategies – kinds and structure of interviews – selection, stress and appraisal interviews.

B. Seminar skills and Presentation skills Strategies-preparation and techniques –telephone and email etiquette- Use of Power point—Techniques.

Chapter -2 Listening and Writing for Career Purposes Skills

Listening- for information and content- Kinds of listening- Factors affecting listening and techniques to overcome them- retention of facts, data and figures- Role of speaker in listening. Types and purposes- Writing business reports, and business proposals- Memos, minutes of meetings- Circulars, persuasive letters- Letters of complaint- ; language and formats used for drafting different forms of communication. Internal and external communication. Leadership Communication Styles - Business leadership – Aspects of leadership-qualities of a leader-training for leadership-delegation of powers and ways to do it-humour-commitment.Identity Crisis in the Corporate – Need for Values, professional and corporate ethics and Social concern – Corporate Social Responsibility –Communication in a crisis -- Knowledge Management – Communication to the public & the mass – Role of Communication in enhancing organization’s health &wealth.

Chapter- 3 Corporate responsibility

Circulating to employees’ vision and mission statements- ethical practices- Human rights -Labour Rights-Environment- governance- Moral and ethical debates surrounding -Public Relations - Building trust with stakeholders. Corporate Ethics and Business Etiquette- Integrity in communication-Harmful practices and communication breakdown- Teaching how to deal with tough clients

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through soft skills. Body language- Grooming- Introducing oneself- Use of polite language- Avoiding grapevine and card pushing – Etiquette in e-mail, mobile and telephone.

References:

1. Corporate Communication. Paul Argenti.2012.
2. Essentials of Corporate Communication: Implementing Practices for Effective Reputation Management. Cees B.M. Van Riel and Charles J Frombrun. Routledge.2007.
3. Corporate Communications: Theory and Practice. Joep Cornelissen. SAGE. 2004.
4. Fundamentals of Corporate Communications. Richard Dolphin and David Reed.Routledge.2009
5. Lesikar's Basic Business Communication - 7th Edition Authors: Raymond V. Lesikar, John D. Pettit, Marie E. Flatley Publisher: Irwin. 1996.
6. Developing Communication Skills -Author(s)Krishna Mohan, Meera Banerji Publisher: Macmillan Publishers India, 2 nd edition 2009.
7. Business Correspondence and Report Writing – Author : R.C. Sharma & Krishna Mohan – 3 rd Edition. Publisher: Tata McGraw-Hill. – 2008

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EE3701 **CONTROL SYSTEMS LAB**

Externals: 60Marks	**L-T-P-C**
Internals: 40Marks	**0-0-2-1**

Course Objectives: To provide the practical exposure

- To strengthen the knowledge of Feedback control To inculcate the controller design concepts
- To familiarize with control systems components like servomotor, synchros
- To familiarize programmable logic controller
- To strengthen the knowledge of Magnetic amplifier**.**

Course Outcomes: At end of this course student will be able to

- Demonstrate time response of second order system
- Understand characteristics of control system components like servomotor, synchros
- Design and understand PID control for temperature control application Design controller and compensators using simulation tools
- Understand characteristics of control system components like Magnetic amplifier

Any Eight of the following experiments are to be conducted:

1. Time response of Second order system

2. Characteristics of Synchro transmitter and receiver

3. Programmable logic controller Study and verification of truth tables of logic gates, simple Boolean expressions and application of speed control of motor.

4. Effect of feedback on DC servo motor

5. Transfer function of DC motor

6. Effect of P, PD, PI, PID Controller on a second order systems

7. Lag and lead compensation Magnitude and phase plot

8. Temperature controller using PID

9. Characteristics of magnetic amplifiers

10. Characteristics of AC servo motor

Any two simulation experiments are to be conducted:-

1. Simulation of Op-Amp based Integrator and Differentiator circuits.

2. Linear system analysis (Time domain analysis, Error analysis) .

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3. Stability analysis (Bode, Root Locus, Nyquist) of Linear Time Invariant system

4. State space model for classical transfer function Verification

REFERENCE BOOKS:

1. Simulation of Electrical and electronics Circuits using PSPICE by M.H.Rashid, M/s PHI Publications.

2. PSPICE A/D manual Microsim, USA.

3. PSPICE reference guide Microsim, USA.

4. MATLAB and its Tool Books manual and Mathworks, USA.

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EE3703 MICROPROCESSORS & MICRO CONTROLLERS LAB

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **0-0-2-1**

Course Objective:

- To familiarize with arduino board and micro controller.
- To understand the interfacing with interfacing with various modules.

Course Outcome:

- The student will be able to interface Keyboard, stepper motor and DC motor.
- The student will be able to interface traffic light, ADC and DAC.
- The student will be able to understand 8051 Microcontroller concepts, architecture, programming and application of Microcontrollers

LIST OF EXPERIMENTS:

1. Interface simple seven segment LED display with arduino and controller.
2. To Display “DEPT OF EEE” on LCD in 8-bit as well as 4-bit mode
3. Interface Keyboard and LCD with controller.
4. Interface Stepper Motor by controlling its direction and make it spin faster or slower arduino and controller..
5. Interface DC motor and control its speed using PWM technique. arduino and controller.
6. Interface Elevator to arduino and controller..
7. Interface Traffic Light with arduino and controller..
8. Interfacing ADC to Microcontroller.
9. Interface DAC with Microcontroller and generate multiple waveforms.
10. Interface Temperature Sensor to ADC and measure it on LCD with arduino and controller.

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THIRD YEAR (E3) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE3201	Power Semiconductor Drives	PCC	4	0	0	4	4
2	EE3202	Power Systems Operationand Control	PCC	3	1	0	4	4
3	EE3801	Power Systems Lab	PCC	0	0	2	2	1
4	EE3803	Electrical Simulation Lab	PCC	0	0	3	3	1.5
5	EE311X	Program Elective-I	PEC	3	0	0	3	3
6	EE322X	Program Elective-II	PEC	3	0	0	3	3
7	EE323X	Program Elective-III	PEC	3	0	0	3	3
8	EE3902	Mini Project	SIP	0	0	2	2	1
9	CS3401	Open Elective-II(Oops)	OEC	3	0	0	0	3
		Total		19	1	9	26	23.5

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POWER SEMICONDUCTOR DRIVES

EE3201

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **4-0-0-4**

Course Objectives: This course will develop students' knowledge in/on

- The fundamentals and dynamics of electric drives
- The various types of the rectifier control and chopper control DC drives
- The AC voltage control, frequency control and slip power recovery control of Induction motor drives.
- Various types of synchronous motor drives and its speed torque characteristics

Course Outcomes: At the end of the course the student will be able to:

- Understand the fundamentals and dynamics of electric drives
- Develop the rectifier control and chopper control DC drives
- Realize the Concept of AC voltage control, frequency control and slip power recovery control of induction motor drives & Solve Problems
- Know the concept of Synchronous motor drives & Solve Problems

UNIT I: Fundamentals of Electric Drives(6 hours)

Electric Drives, advantages of electric drives, parts of electric drives, choice of electric drives, status of D.C. drives and A.C. drives. Starting, Braking, speed control of AC and DC motors.

UNIT II: Dynamics of Electric drives (8 hours)

Fundamental torque equations, types of load, Quadrant diagram of speed-Torque characteristics, Dynamics of load torque combinality, steady state stability and Transient stability of an Electric drives. Load equalization. Calculation of time and energy loss in Transient operation, Drive specifications.

UNIT III: Control of DC drives (10 hours)

Controlled rectifier circuits, braking operation of rectifier controlled separately excited dc motor, single phase and three phase half and fully controlled rectifier fed separately excited dc motor, multi quadrant operation of fully controlled rectifier fed separately excited dc motor.
Chopper control of DC drives: chopper control of separately excited and series dc motors , multi quadrant control of chopper fed motors with an examples.

UNIT IV: Control of Induction Motor Drives: (10 hours)
Control of induction motor by AC voltage controllers, Frequency controlled Induction motor drives: control of Induction motor by Voltage Source Inverter (VSI), Current controlled PWM inverters and cyclo converters, Slip power controlled wound-rotor induction motor drives, static rotor resistance control, static scherbius drives, krammers drives.

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UNIT V: Control of Synchronous Motor Drives (8 Hours)

Operation of cylindrical rotor synchronous motor from VSI and CSI, self controlled Synchronous Motor Drives using cyclo converters, Permanent magnet AC motor drives.

Text Books:

1. G.K. Dubey, "Fundamentals of Electrical Drives", Narosa Publishers, New Delhi. 1988
2. N.K. De and P.K. Sen, "Electrical Drives", Prentice Hall of India, New Delhi. 1999

References:

1. Vedam Subrahmanyam, "Thyristor Control of Electrical Drives", Tata McGraw Hill, New Delhi. 1988.
2. B.K. Bose "Modern Power Electronics & A.C Drives'.Pearson .edu
3. P.S.Bimbhra " Power Electronics" Khanna publishers

G.K. Dubey, "Power Semiconductor Drives", Narosa Publishers, New Delhi. 1988

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EE3801 **POWER SYSTEMS LAB**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **0-0-2-1**

Course Objectives:

- Performance of long transmission lines and reactive power control
- Characteristics of protective relays
- Short circuit analysis and sequence components of power system elements
- Study of different faults on Transmission lines.

Course Outcomes: After completion of this lab, students will be able to

- Determine the performance characteristics of a long transmission line
- Determine the performance of reactive power control
- Determine the operating characteristics of protective relays
- Compute fault currents and determine the sequence components of power system elements.

List of experiments

1. Determination of Sequence Impedances of a cylindrical rotor Synchronous Machine.
2. Determination of Positive, Negative and zero sequence reactance of 3 phase Transformers
3. Fault analysis of 3 phase Alternator, (LG, LL, LLG, LLLG faults).
4. Determination of Sub transient reactance's of a Salient Pole Synchronous Machine.
5. To obtain the operating characteristics of IDMT over current relay
6. Characteristics of Percentage biased of Static Differential Relay
7. Performance and Testing of Generator Protection System
8. To obtain the performance characteristics of long transmission line
9. To determine the breakdown strength of oil
10. Reactive power control of long Transmission line

Any two simulation experiments listed below should be conducted using two electrical related software

1. Distribution System Reliability Analysis.
2. Power System Fault Analysis.
3. Transmission Line Fault Analysis

Text Books:

1. C.L. Wadhwa: Electrical Power Systems – New Age International Pub.Co.Third Edition, 2001.
2. Badriram and D.N. Vishwakarma, Power System Protection and Switchgear, TMH 2001

References:

1. D.P. Kothari and I.J. Nagrath, Modern Power System Analysis - Tata Mc Graw Hill Pub. Co., New Delhi, Fourth edition, 2011 Reference.
2. Switchgear and Protection – by Sunil S Rao, Khanna Publishers, 1992.
3. "Electrical Power", by S. L. Uppal, Khanna pulishers,1988.

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EE3803 ELECTRICAL SIMULATION LAB

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **0-0-3-1.5**

Course Objective:

- To impart hands on experience in verification of circuit laws and theorems, measurement of circuit parameters, study of circuit characteristics and the simulation of power electronics circuits using PSIM & MATLAB.
- Gives practical exposure to the usage of different circuits with different condition.
- Acquire skills of using computer packages PSIM, POWER WORLD and MATLAB coding and SIMULINK in power Electronics and power system studies.

Course Outcome:

Upon the successful completion of this course, the student is expected to gain the following skills:

- Understand the fundamentals and programming Knowledge in MATLAB.
- Able to understand the Transient & Steady State Performance of a system.
- Able to generate plots and export this for use in reports and presentations.
- Able to give practical experience with simulating physical systems

List of Experiments:

1. Stability analysis (Bode, Root locus, Nyquist) of linear time invariant system using MATLAB.
2. Effect P, PD, PI, PID controllers on a second order system using MATLAB.
3. Simulation of Single Phase Semi & Full bridge rectifier using PSIM.
4. Simulation of three semi converter using PSIM.
5. Simulation of three full bridge rectifier using PSIM.
6. Simulation of single phase bridge inverter using PSIM.
7. Simulation of DC-DC Converter using PSIM.
8. Simulation of three phase inverter 120 Degree mode using MATLAB programming.
9. Simulation of three phase inverter 180 Degree mode using MATLAB programming.
10. Load curve analysis using MATLAB programming.
11. Performance evaluation of long and short transmission line using MATLAB programming.
12. Newton Raphson method of load flow analysis using Power World.
13. Gauss Seidal method of load flow analysis using Power World.
14. Fault analysis using Power World.
15. Modelling of DC motor using MATLAB.

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EE3202 POWER SYSTEMS OPERATION AND CONTROL

Externals: 60Marks **L-T-P-C**

Internals: 40 Marks **3-1-0-4**

Course Objective: At the end of the course the student will be able to:

- To understand the computation of load flows in a power systems
- To study the various methods of reactive power control in power systems and economic
- load scheduling
- To study load frequency control and its analysis in an isolated power system
- To study stability, stability limits and the dynamics of synchronous machines

Course Outcomes: After completion of this course, students will be able to:

- Compute the bus variables and the power flows in the system using various
- iterative methods
- Determine the optimal economic load scheduling.
- Determine the static and dynamic frequency response of a power system for a single
- area and two area system
- Predict the stability of power systems and determine the transient stability limits

UNIT I: Load flow studies (8 hours)

Introduction, Bus classification, Nodal admittance matrix, Transmission Network Representations: Bus Admittance frame and Bus Impedance frame. Formation of Ybus: Direct and Singular Transformation Methods, Load flow equations, Iterative methods – Gauss, Gauss Seidel and Newton Raphson methods. Newton decoupled and fast decoupled. Merits and Demerits of these methods, system data for load flow study.

UNIT II : Economic Operation of Power Systems (6 hours)

Distribution of load between units within a plant, transmission loss as a function of plant generation, calculation of loss coefficients, distribution of load between plants. Unit commitment: Introduction, constraints in unit commitment problems.

UNIT III: Load Frequency control (11 hours)

Introduction, Load frequency problem, Megawatt frequency (or P-F) control channel, Megavar voltage (or Q – V) control channel. Dynamic interaction between P-F and Q-V loops, Mathematical model of speed governing system, turbine models division of power system into control areas, P-F control of single control area (the uncontrolled and controlled cases) P-F control of two area systems (the uncontrolled and controlled cases).

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UNIT IV: Power System Stability (8 hours)

The stability problem, steady state stability limit, Expression using ABCD parameters, steady state stability of synchronous machine. transient stability, swing equation, equal area criterion of stability and its further applications, step by step solution swing equation, some factors affecting transient stability & Methods of improving stability . Concept of Dynamic stability – effect of excitation on generator power limits.

UNIT V: Reactive Power–Voltage Control (9 hours)

Basics of reactive power control. Excitation systems – modeling. Static and dynamic analysis - stability compensation - generation and absorption of reactive power. Relation between voltage, power and reactive power at a node - method of voltage control - tap-changing transformer. System level control using generator voltage magnitude setting, tap setting of OLTC transformer and MVAR injection of switched capacitors to maintain acceptable voltage profile and to minimize transmission loss.

Textbooks:

1. John Grainger & William Stevenson Jr., “Power Systems Analysis”, McGraw Hill, 1/e,

2. D.P.Kothari and I.J.Nagrath, Modern Power System Analysis, 4th Edn, Tata McGraw Hill Education Private Limited 2011.

3. C.L.Wadhwa, Electrical Power Systems, 3rd Edn, New Age International Publishing Co., 2001.

Reference Books:

1. Olle I Elgerd “ Electric Energy Systems Theory”, Tata McGraw Hill ,2/e ,2011

2. Chakrabarthi, Abhijit halder, “Power system analysis: Operation and Control”, Prentice hall of India, 3/e, 2010.

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CS3401 OBJECTED ORIENTED PROGRAMMING

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Objectives:

- To understand Object oriented programming concepts, and apply them in Problem solving.
- To learn the basics of Java Console and GUI based programming

UNIT-1: Introduction to OOPS

Paradigms of Programming Languages, Basic concepts of Object Oriented Programming, Differences between Procedure Oriented Programming and Object Oriented Programming, Objects and Classes, Data abstraction and Encapsulation, Inheritance, Polymorphism, benefits of OOP , application of OOPs.

Java: History, Java features, Java Environment, JDK, and API.

Introduction to Java : Types of java program, Creating and Executing a Java program, Java Tokens, Keywords, Character set, Identifiers, Literals, Separator, Java Virtual Machine (JVM), Command Line Arguments, Comments in Java program.

UNIT -2: Elements

Constants, Variables, Data types, Scope of variables, Type casting,

Operators: Arithmetic, Logical, Bit wise operator, Increment and Decrement, Relational, Assignment, Conditional, Special operator, Expressions – Evaluation of Expressions Decision making and Branching: Simple if statement, if, else statement, Nesting if, else, else if Ladder, switch statement, Decision making and Looping: While loop, do-While loop, for loop, break, labelled loop, continue Statement, Simple programs

Arrays: One Dimensional Array, Creating an array, Array processing, Multidimensional Array, Vectors, Wrapper classes, Simple programs

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UNIT-3: Strings:

Exploring String class, String Class Methods, String Buffer Class, Simple programs Class and objects: Defining a class, Methods, Creating objects, Accessing class members, Constructors, Static members, Nesting of Methods, this keyword, Command line input. Polymorphism – Static Polymorphism, Dynamic Polymorphism, Method overloading, Polymorphism with Static Methods, Private Methods and Final Methods.

Inheritance: Defining a sub class, Deriving a sub class, Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Overriding methods, Final variables and methods, Final classes, Finalizer methods, Abstract methods and classes, Visibility Control: Public access, Private access, default and protected. Abstract classes, Interfaces - Interfaces vs Abstract classes, defining an interface, implementing interfaces, accessing implementations through interface references, extending interfaces. Inner classes - uses of inner classes, local inner classes, anonymous inner classes, static inner classes, examples.

UNIT- 4: Packages

Java API Packages, System Packages, Naming Conventions, Creating & Accessing a Package, Adding Class to a Package, Hiding Classes, Programs Exception Handling: Limitations of Error handling, Advantages of Exception Handling, Types of Errors, Basics of Exception Handling, try blocks, throwing an exception, catching an exception, finally statement Multi threading: Creating Threads, Life of a Thread, Defining & Running Thread, Thread Methods, Thread Priority, Synchronization, Implementing runnable interface, Thread scheduling.

UNIT-5: AWT Components and Event Handlers

Abstract window tool kit, Event Handlers, Event Listeners, AWT Controls and Event Handling: Labels, Text Component, Action Event, Buttons Check Boxes, Item Event, Choice, Scrollbars, Layout Managers- Input Events, Menus, Programs GUI Programming with Java - Introduction to Swing, limitations of AWT, Swing vs AWT, MVC architecture, Hierarchy for Swing components, Containers – J Frame, J Applet, J Dialog, J panel. Overview of some swing components J button, J Label, J Text Field, J text Area, simple swing applications. I/O Streams: File, Streams, Advantages, The stream classes, Byte streams, Character streams. JDBC, ODBC Drivers, JDBC ODBC Bridges, Seven Steps to JDBC, Importing java SQL Packages, Loading &

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Registering the drivers, Establishing connection. Creating & Executing the statement.

TEXT BOOKS:

1. Java the complete reference, 7 th editon, Herbert Schildt, TMH.

2. Understanding OOP with Java, updated edition, T. Budd, Pearson Eduction.

REFERENCE BOOKS:

1. An Introduction to programming and OO Design using Java, J.Nino and F.A. Hosch, John wiley & Sons.

2. Introduction to Java Programming, Y. Daniel Liang, Pearson Education

3. An Introduction to Java programming and Object Oriented Application Development, R.A. Johnson-Thomson

4. Programming with Java - E. Balagurusamy

5. Object oriented Programming in Java - Dr. G.Thampi

6. Let us Java – Yashavant Kanetkar - BPB Publications, New Delhi - First Edition 2012

7. Core Java, An Integrated Approach, Dr. R. Nageswara Rao

8. An Introduction to Oops with Java - C Thomas WU - TataMc-Graw Hill, New Delhi - 4th Edition

9. Object oriented Programming through Java - ISRD Group - TataMc-Graw Hill, New Delhi - Eight Reprint 2011

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FOURTH YEAR (E4) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	EE4101	Utilization of Electrical Energy	PCC	3	0	0	3	3
2	EE4701	Energy Systems Lab	PCC	0	0	2	2	1
3	EE414X	Program Elective-IV (ANN)	PEC	3	0	0	3	3
4	EE415X	Program Elective-V (HEV)	PEC	3	0	0	3	3
5	EE4901	Major Project-I	SIP	0	0	8	8	4
6	EE4900	Summer Internship	SIP	0	0	0	0	1
7	HS3201	Communication Skills	HSMC	1	0	0	1	1
8	BM4107	MEFA	HSMC	3	0	0	3	3
12	EE4000	Comprehensive Viva	SIP	0	0	0	0	1
Total				13	0	10	23	20

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EE4101 **UTILIZATION OF ELECTRICAL ENERGY**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives: This course will develop knowledge in/on

- Various electric traction systems with their performance.
- Selection of motor for different industrial drives.
- Electric heating and welding techniques.
- Designing and selection of lamps for proper illumination

Course Outcomes: After completion of this course, students will be able to

- Choose the motor for different types of Electric traction systems.
- Evaluate the selection of a motor for different types of loads.
- Use various heating and welding techniques for different applications.
- Select and design the lamps for proper illumination

UNIT I: INDUSTRIAL UTILIZATION (8 hours)

Introduction, Factors governing selection of Electric Motors, Nature of electric supply, Types of drives, Nature of loads, Standard Ratings of Motors, Choice of ratings of Motors, Types of Motors used in industrial Drives, speed control.

UNIT II : ELECTRIC HEATING and WELDING (8 Hours)

Advantages and methods of electric heating, resistance heating, induction heating and dielectric heating. Electric welding, resistance and arc welding, electric welding equipment, comparison between A.C. and D.C. Welding.

UNIT III: ILLUMINATION SYSTEMS (10 Hours)

Understanding various terms regarding light, lumen, intensity, candle power, lamp efficiency, specific consumption, glare, space to height ratio, waste light factor, depreciation factor, various illumination schemes, Incandescent lamps and modern luminaries like CFL, LED and their operation, energy saving in illumination systems, design of a lighting scheme for a residential and commercial premises, flood lighting, Discharge lamps

UNIT IV: ELECTRIC TRACTION (10 Hours)

System of electric traction and track electrification, Review of existing electric traction systems in India, Special features of traction motor, methods of electric braking-plugging rheostat braking and regenerative braking, Mechanics of train movement, Speed-time curves for different services trapezoidal and quadrilateral speed time curves, Calculations of tractive effort, power, specific energy consumption for given run, effect of varying acceleration and braking retardation, adhesive weight and coefficient of adhesion.

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UNIT V: RESIDENTIAL ELECTRICAL SYSTEMS (8 Hours)

Types of residential wiring systems, general rules and guidelines for installation, load calculation and sizing of wire, rating of main switch, distribution board and protection devices, earthing system calculations

Introduction to Electric Vehicles: Historical Journey of Hybrids and Electric Vehicle, Economic and Environmental Impact of Electric Hybrid Vehicle

Text Books:

1. E. Openshaw Taylor, Utilisation of Electric Energy by University press.
2. C.L. Wadhwa, Generation, Distribution and Utilization of electrical Energy, New Age International (P) Limited, Publishers, 1997

Reference Books:

1. N.V.Suryanarayana, Utilization of Electrical Power including Electric drives and Electric traction, New Age International (P) Limited, Publishers, 1996.
2. Partab, Art & Science of Utilization of electrical Energy Dhanpat Rai & Sons.

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EE4701 **ENERGY SYSTEMS LAB**

Externals: 60Marks L-T-P-C

Internals: 40Marks 0-0-2-1

Course Objective:

- To introduce the basic renewable energy systems
- To study the V-I characteristics of solar panel
- To study the wind energy system
- To study the MPPT Techniques

Course Outcome:

- The student will be able to understand PV cell characteristics
- Understand the characteristics of solar panal
- To simulate the P-V and DC-DC converter
- To solve Fuel cell and Battery concepts
- The student will be able understand to interface with power converters and MPPT concept

List of Experiments:

1. V-I characteristics of solar panel at various levels of isolation.

2. Study of wind turbine generator.

3. Performance Study of Solar Flat Plate Thermal Collector Operation with Variation in Mass Flow Rate and Level of Radiation

4. Characterization of Various PV Modules Using large area Sun Simulator

5. Study of micro-hydel pumped storage system

6. Fuel Cell Experiment

7. Study of 100 kW solar PV plant

8. Simulation of PV and DC-DC converter interface

9. Simulation of MPPT for PV cell or module

10. Simulation of PV cells in parallel and series

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HS3201 Communication Skills

Contact Hours per Week 40 **L- T- P -C**

Internal External Credits 60 **1- 0- 0 -1**

Course Objectives

- To make the students efficient communicators via experiential learning.
- To enhance learners' analytical and creative skills, so that they will be capable of addressing a wide variety of challenges in their professional lives.
- To help learners to improve the leadership qualities and professional etiquette
- To expose learners to an effective communicative environment.

Course Outcomes

- Develop interpersonal communication, small group interactions and public speaking.
- Develop confidence and skills related to reading comprehension.
- Improve a logical framework for the critical analysis of spoken, written, visual and mediated messages upon diverse platforms.
- Demonstrate the ability to apply vocabulary in practical situations.

UNIT-I: Introduction to communication

Introduction – Importance of Communication Skills – Definition – Scope and Nature – Verbal and Nonverbal communication.

UNIT-II: Reading Skills Reading Comprehension of unseen passage

Prose – News Paper Reading and Analysis (Editorial), Novels, different research articles, crack the answers in the unseen paragraphs in the competitive exams etc….

UNIT-III: Functional Grammar

1. Parts of Speech (Functional Usage)
2. Subject and predicate (useful at workstations)
3. Conjunctions-Gap fillers (Linkers; connectors; cohesive devices)
4. Verbs & Verb patterns (Transitive and Intransitive - Finite and Infinite - Regular and Irregular –Modals)
5. Tenses (Various Applications)
6. Prepositions/ Prepositional verbs (idiomatic expressions, one word substitutions, phrasal verbs)
7. Adjectives (describing, narrating, effective presentation)

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UNIT-IV: Vocabulary
Enhancing Vocabulary, Developing Professional vocabulary, different forms of verbs (verb forms: v1, v2,v3) – Using Dictionary: Spelling – jargon specific vocabulary, context specific vocabulary, right choiceof vocabulary etc…

UNIT-V: Composition Paragraph
Essay - Expansion - Describing the Pictures – Giving Directions – Situational

Dialogue Writing – Social and Professional Étiquette – Telephone Etiquette, email etiquettes, Role plays, JAM, and elocution etc…

Text Books
1. Communication skills by M. Raman and Sangeeta Sharma

References
1. Joseph Mylal Biswas book of English Grammar
2. R. Murphy - Cambridge Press
3. Wren and Martin
4. The Good Grammar book by OUP
5. How to Win Friends and Influence people by Dale Carnigie
6. How to Read and Write Better by Norman Lewis
7. Better English by Norman Lewis
8. Use of English Collocations by OUP
9. www.humptiesgrammar.com
10. www.bbcenglisgh.com
11. www.gingersoftware.com
12. www.pintest.com

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BM4107

MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTACY

Internal Assessment 40	L-T-P-C
End Exam 60	3-0-0-3

- Understand the nature and scope of managerial economics and the concepts of demand analysis
- Understand the significance of demand elasticity and the concepts of demand forecasting.
- Understand the concepts of production and cost analysis
- Understand the concepts of production and cost analysis different market structures and their Competitive situations.
- Understand the concept and significance of capital budgeting.

UNIT-1 Introduction to Managerial Economic

Definition-Nature and Scope-Basic economic principles -the concept of Opportunity cost, Marginalism, Incremental concept, Time perspective, Discounting principle, Risk and Uncertainty.

UNIT- 2 Theory of Demand and Supply

Demand Analysis -Demand function- Law of demand- Determinants of demand and Types of demand- Elasticity of demand-Types- Demand Forecasting,-Need for Demand Forecasting-Methods of Demand Forecasting -Supply - Law of supply.

UNIT-3 Theory of production & cost analysis

Production-Meaning, Production function-Production function with one variable- Production function with two variables- Isoquants and Isocosts -Marginal Rate of Technical Substitution-Returns to Scale- Cost concepts: Meaning of Costs- Types of Costs.

UNIT-4 Market Structure and Pricing Practices

Classification of Market Structures -Features -Competitive situations-Price-Output determination under Perfect competition- Monopoly, Features of Monopolistic competition and Oligopoly-Pricing Strategies.

UNIT-5 Capital and Capital Budgeting

Capital-Introduction of Capital- Definition of Capital-Sources of Capital.

Capital Budgeting: Significance of Capital Budgeting- Need for capital budgeting decisions-Capital Budgeting decisions- Kinds of capital budgeting Decisions- Methods of Capital

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Budgeting-Traditional Methods-Payback period and Accounting rate of return methods, Discounted Cash flow methods- Net present value method.

Text Books:

1. A.R. Aryasri, Managerial Economics and Financial Analysis, McGraw Hill Education

2. Varshiney and Maheswari, Managerial Economics, Sultan Chand & Co, New Delhi

Reference Books:

1. Vanita Agarwal, Managerial Economics, Pearson Education

2. Domnick Salvatore: Managerial Economics in a Global Economy, 4th Edition, Thomson

3. S.P. Jain and K.L. Narang, Financial Accounting

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FOURTH YEAR (E4) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	XXXXX	Open Elective-III	OEC	3	0	0	3	3
2	XXXXX	Open Elective-IV	OEC	3	0	0	3	3
3	EE4902	Major Project-II	SIP	0	0	12	12	6
Total				6	0	12	12	12

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Programme Elective Courses (PECs)

S. No	Course Code	Course Title	Semester	Course Category	Hours per week			Total Contact Hours	Credits
					L	T	P		
1	EE3211	Power System Protection	VI	PEC	3	0	0	3	3
	EE3212	Reliability Engineering	VI	PEC	3	0	0	3	3
	EE3213	Special Electrical Machines	VI	PEC	3	0	0	3	3
	EE3214	Advanced Topics in Power Electronics	VI	PEC	3	0	0	3	3
	EExxxx	MOOC Course	VI	PEC	3	0	0	3	3
2	EE3221	Wind and Solar Energy Systems	VI	PEC	3	0	0	3	3
	EE3222	Digital Control Systems	VI	PEC	3	0	0	3	3
	EE3223	Electrical Distribution Systems	VI	PEC	3	0	0	3	3
	EE3224	Switched Mode Power Supplies	VI	PEC	3	0	0	3	3
	EExxxx	MOOC Course	VI	PEC	3	0	0	3	3
3	EE3231	HVDC Transmission	VI	PEC	3	0	0	3	3
	EE3232	VLSI Design	VI	PEC	3	0	0	3	3
	EE3233	Digital Signal Processing	VI	PEC	3	0	0	3	3
	EE3234	Design of Photovoltaic Systems	VI	PEC	3	0	0	3	3
	EEXXXX	MOOC Course	VI	PEC	3	0	0	3	3
4	EE4141	Smart Grid	VII	PEC	3	0	0	3	3
	EE4142	Artificial Neural Networks and Fuzzy Systems	VII	PEC	3	0	0	3	3
	EE4143	Flexible AC Transmission System	VII	PEC	3	0	0	3	3
	EE4144	IOT in Micro grid	VII	PEC	3	0	0	3	3
	EEXXXX	MOOC Course	VII	PEC	3	0	0	3	3
5	EE4151	Power Quality	VII	PEC	3	0	0	3	3
	EE4152	Industrial Automation and Control	VII	PEC	3	0	0	3	3
	EE4153	High Voltage Engineering	VII	PEC	3	0	0	3	3
	EE4154	Hybrid Electrical Vehicle	VII	PEC	3	0	0	3	3
	EEXXXX	MOOC Course	VII	PEC	3	0	0	3	3

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EE3211 **POWER SYSTEMS PROTECTION**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objective:

- To compare and contrast electromagnetic, static and microprocessor based relays
- To apply technology to protect power system components
- To select relay settings of over current and distance relays.
- To analyze quenching mechanisms used in air, oil and vacuum circuit breakers

Course Outcomes: After completion of this course, students will be able to

- Compare electromagnetic with static relays
- Evaluate the of performance of Various Relays
- Understand about the concept of over voltage protection and insulation coordination
- Analyze Fundamental principles of circuit breakers & fuses

UNIT I: Relays (8 Hours)

Electromagnetic Relays - Basic Requirements of Relays – Primary and Backup Protection – Construction, Details of – Attracted Armature, Balanced Beam, Inductor Type and Differential Relays – Universal Torque Equation – Characteristics of Over Current, Direction and Distance Relays. Static Relays –Introduction to static relays, Advantages and Disadvantages over electromagnetic relays

UNIT II: Protection of Generators and Transformers (10 Hours)

Protection of Generators against Stator Faults, Rotor Faults, and Abnormal Conditions. Restricted Earth Fault and Inter-Turn Fault Protection. Numerical Problems On percentage Winding Unprotected. Protection of Transformers: Percentage Differential Protection, Numerical Problem on Design ofCT s Ratio, Buchholtz Relay Protection, Numerical Problems.

UNIT III: Protection of Transmission Lines (8 Hours)

Protection of Feeder (Radial & Ring Main) Using Over Current Relays. Protection of transmission Line– 3 Zone Protection Using Distance Relays. Carrier Current Protection. Protection of Bus Bars.

UNIT IV: Overvoltage Protection and Insulation Coordination (8 Hours)

Over voltage due to arcing ground and Peterson coil, lightning, horngaps, surge diverters, rod gaps, expulsion type lightning arrester, valve type lightning arrester, ground wires, ground rods, counter poise, surge absorbers, insulation coordination, volt-time curves.

UNIT V: Circuit Breakers (10 Hours)

Introduction, arcing in circuit breakers, arc interruption theories, re-striking and recovery voltage, resistance switching, current chopping, interruption of capacitive current, oil circuit breaker, air blast circuit breakers, Vacuum Circuit Breaker, SF6 circuit breaker, operating mechanism, selection of circuit breakers, high voltage d.c. breakers, ratings of circuit breakers, testing of circuit breakers. Fuses : Introduction, fuse characteristics, types of fuses, application

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of HRC fuses, discrimination

Text Books:

1. Badriram and D.N. Vishwakarma, Power System Protection and Switchgear, TMH 2001
2. U.A.Bakshi, M.V.Bakshi: Switchgear and Protection, Technical Publications, 2009.
3. Switchgear and Protection – by Sunil S Rao, Khanna Publishers, 1992.

Reference

1. L.P.Singh ―Protective relaying from Electromechanical to Microprocessors‖, New Age International
2. "Electrical Power", by S. L. Uppal, Khanna pulishers,1988.

Ravindranath & Chander, "Switch Gear & Protection" New Age International , 2/e,

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EE3212 **RELIABILITY ENGINEERING**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives:

- The course covers principles of reliability, failure rate and its relation to reliability, probability distribution of the time to failure, exponential and weibull distributions, reliability of systems, series and parallel systems, stand by redundancy, systems mean time to failure, mean residual life, reliability in design.
- It also includes failure mode effect analysis, failure tree analysis, reliability testing and analysis, And warranty problems

Course Outcomes: At the end of the course the student will be able to:

- Have a working knowledge of the techniques of reliability engineering
- To apply learned concepts to improving the maintenance, the maintainability, hazard risk and the safety of a plant.
- Develop warranty plans for different products
- To carry out a failure mode effect and criticality analysis.

UNIT I: Basic Probability Theory(9 Hours)

Elements of probability, probability distributions, Random variables, Density and Distribution functions- Binomial distribution- Expected value and standard deviation - Binomial distribution, Poisson distribution, normal distribution, exponential distribution, Weibull distribution.

Definition of Reliability
Definition of terms used in reliability, Component reliability, Hazard rate, derivation of the reliability function in terms of the hazard rate. Hazard models - Bath tub curve, Effect of preventive maintenance. Measures of reliability: Mean Time to Failure and Mean Time Between Failures.

UNIT II: Network Modeling And Evaluation Of Simple Systems(8 Hours)

Basic concepts- Evaluation of network Reliability / Unreliability - Series systems, Parallel systems - Series-Parallel systems- Partially redundant systems- Examples.
Network Modeling and Evaluation of Complex systems
Conditional probability method- tie set, Cutset approach- Event tree and reduced event tree methods-Relationships between tie and cutsets- Examples.

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UNIT III: Time Dependent Probability(8 Hours)

Basic concepts- Reliability function f(t). F(t), R(t) and h(t) - Relationship between these functions.

Network Reliability Evaluation Using Probability Distributions

Reliability Evaluation of Series systems, Parallel systems – Partially redundant systems- determination of reliability measure- MTTF for series and parallel systems – Examples.

UNIT IV: Discrete Markov Chains(6 Hours)

Basic concepts- Stochastic transitional probability matrix- time dependent probability evaluation-Limiting State Probability evaluation- Absorbing states –Examples

Continuous Markov Processes

Modeling concepts- State space diagrams- Unreliability evaluation of single and two component repairable systems

UNIT V: (8 Hours)

Frequency And Duration Techniques: Frequency and duration concepts, application to multi state problems, Frequency balance approach.

Approximate System Reliability Evaluation: Series systems – Parallel systems- Network reduction techniques- Cut set approach- Common mode failures modeling and evaluation techniques- Examples.

TEXT BOOKS

1. Roy Billinton and Ronald N Allan, Reliability Evaluation of Engineering Systems, Plenum Press.
2. E.Balagurusamy, Reliability Engineering by Tata McGraw-Hill Publishing Company Limited

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EE3213 **SPECIAL ELECTRICAL MACHINES**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objective:

- To understand the working principle and construction of stepper motors and switched reluctance motors.
- Demonstrate the ability to understand the construction of ,Amplidyne ,Metadyne and Regulex- third brush generator
- To gain knowledge in principle of operation and characteristics of permanent magnet
- To impart knowledge on performance of stepping motors

Course Outcome: At the end of the course the student will be able to:

- Identify the different features of special machines.
- Perform and analyze different methods to find Torque Pulsations for 180 degrees pole arc and 120 degree current sheet Brushless dc motors
- Elucidate the working and characteristics of stepper motors, VRSM motor and PMMM motor.
- Ability to select a special Machine for a particular application.

UNIT I : Special Types Of D.C Machines (8 Hours)

Series booster, Shunt booster, Non-reversible booster & Reversible boosterArmature excited machines, Rosenberg generator, Amplidyne and metadyne, Rototrol and Regulex, third brush generator, three-wire generator and dynamometer.

UNIT II: Stepper Motors (9 Hours)

Introduction, synchronous inductor (hybrid) stepper motor, construction, principle of operation, energization with two phase at a time, essential conditions for the satisfactory operation of a 2-phase hybrid step motor, very slow speed synchronous motor for servo control, different configurations for switching the phase windings, control circuits for stepping motors and an open-loop controller for a 2-phase stepping motor.

UNIT III: Variable Reluctance Stepper and Switched Reluctance Motor (10 Hours)

Variable reluctance Stepper motors (VRSM): single-stack VRSM, Multiple stack VRSM, Open-loop control of 3-phase VRSM, closed-Loop control of stepper motor: discriminator (or rotor position sensor) translator, major loop operation; characteristics of stepper motor in open loop drive , comparison

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between open-loop position control with step motor and a position control servo using a conventional (dc or ac) servo motor-, Suitability and areas of application of stepper motors.

Switched Reluctance Motor (SRM): Introduction, improvements in the design of conventional reluctance motors, Some distinctive differences between SR and conventional reluctance motors, principle of operation of SRM, Some design aspects of stator and rotor pole arcs, design of stator and rotor and pole arcs in SR motor, determination of L(θ) & θ profile , power converter for SR motor, A numerical example , Rotor sensing mechanism and logic control, drive and power circuits, position sensing of rotor with Hall problems, derivation of torque expression-general and linear case.

UNIT IV: Permanent Magnet Motors and Brushless DC Motor (10 Hours)

Permanent Magnet Motors: Introduction, Hysteresis loops and recoil line, stator frames (pole andyoke) of conventional PMDC Motors, Equivalent circuit of a PMDC motor, Development of Electronically commutated dc motor from conventional dc motor.

Brushless DC Motor: Types of construction, principle of operation of BLDC motor, sensing andswitching logic scheme, sensing logic controller, lockout pulses, drive and power circuits, Base drive circuits, power converter circuit, Theoretical analysis and performance prediction, modeling and magnet circuit d-q analysis of BLDC motors, transient analysis formulation in terms of flux linkages as state variables, Approximate solution for current and torque under steady state, Theory of BLDC motor as variable speed synchronous motor (assuming sinusoidal flux distribution), Methods or reducing Torque Pulsations, 180 degrees pole arc and 120 degree current sheet.

UNIT V: Linear Induction Motor(LIM) (5 Hours)

Development of a double sided LIM (DSLIM) from rotary type IM, A schematic of LIM drive for electric traction development of one sided LIM with back iron, field analysis of a DSLIM fundamental assumptions.

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TEXT BOOKS

- K.Venkataratnam, Special electrical machines, university press.
- R.K. Rajput - Electrical machines - 5th edition.

Reference Books:

1. A.E.FitzgeraldandC.Kingsley,"ElectricMachinery",NewYork,McGrawHill Education,2013.
2. E.ClaytonandN.N.Hancock,“PerformanceanddesignofDCmachines”,CBS Publishers,2004.
3. M.G. say, ,“PerformanceanddesignofACmachines”, CBSPublishers,200

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EE3214 **ADVANCED TOPICS IN POWER ELECTRONICS**

L-T-P-C

3-0-0-3

Course Objectives

- To understand High efficiency, High reliability, long lifetime, and Fast dynamic response features of advanced power electronic devices.
- To understand Non-linear operation and related losses in the power electronic converters.
- To be aware of application of semiconductor devices in control of electrical energy.
- To study ideal rectifier operation and design and to understand the operation of multi-resonant converters and load resonant converters.

Course Outcomes: After completion of this course, the students shall be able to:

- Understand and simulate control circuits of power electronic converters.
- Get acquaintance with digital logic control concepts required for converter controls.
- Understand advanced converter topologies operation and area of application.
- Solve Non-linear switched mode power converters operation issues and Simulate Multi pulse PWM on current source converters.

UNIT-1: Introduction to switches

Advanced Silicon devices Silicon HV thyristors, MCT, BRT &EST. SIC devices diodes, thyristors, JFETs & IGBTs. Gallium nitrate devices - Diodes, MOSFETs.

UNIT-II: Pulse Width Modulated Rectifiers

Properties of ideal rectifier, realization of near ideal rectifier, control of the current waveform, single phase and three-phase converter systems incorporating ideal rectifiers and design examples. Non-linear phenomena in switched mode power converters, Bifurcation and Chaos.

UNIT-III: Control of DC-DC converters and soft switching

Control of DC-DC converters: State space modeling of Buck, Boost, Buck-Boost, Cuk Fly back, Forward, Push-Pull, Half & Full-bridge converters. Closed loop voltage regulations using state feedback controllers.Soft-switching DC - DC Converters: zero-voltage-switching converters, zero-current switching converters, multi-resonant converters and load resonant converters. servo

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using a conventional (de or ac) servo motor, Suitability and areas of application of stepper motors.

Switched Reluctance Motor (SRM): Introduction, improvements in the design of conventional reluctance motors, Some distinctive differences between SR and conventional reluctance motors, principle of operation of SRM, Some design aspects of stator and rotor pole arcs, design of stator and rotor and pole arcs in SR motor, determination of L(0) &0 profile, power converter for SR motor, A numerical example, Rotor sensing mechanism and logic control, drive and power circuits, position sensing of rotor with Hall problems, derivation of torque expression-general and linear case.

UNIT IV: Permanent Magnet Motors and Brushless DC Motor (10 Hours)

Permanent Magnet Motors: Introduction, Hysteresis loops and recoil line, stator frames (pole andyoke) of conventional PMDC Motors, Equivalent circuit of a PMDC motor, Development of Electronically commutated de motor from conventional de motor.

Brushless DC Motor: Types of construction, principle of operation of BLDC motor, sensing andswitching logic scheme, sensing logic controller, lockout pulses, drive and power circuits, Base drive circuits, power converter circuit, Theoretical analysis and performance prediction, modeling and magnet circuit d-q analysis of BLDC motors, transient analysis formulation in terms of flux linkages as state variables, Approximate solution for current and torque under steady state, Theory of BLDC motor as variable speed synchronous motor (assuming sinusoidal flux distribution), Methods or reducing Torque Pulsations, 180 degrees pole arc and 120 degree current sheet.

UNIT V: Linear Induction Motor(LIM) (5 Hours)

Development of a double sided LIM (DSLIM) from rotary type IM, A schematic of LIM drive for electric traction development of one sided LIM with back iron, field analysis of a DSLIM fundamental assumptions.

Electrical and Electronics Engineering Dept.

TEXT BOOKS

1. K. Venkataratnam, Special electrical machines, university press.

2. R.K. Rajput - Electrical machines - 5th edition.

3. V.V. Athani International publishers. Stepper motor: Fundamentals, Applications and Design, New age international publishers.

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EE3221 **WIND AND SOLAR ENERGY SYSTEMS**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-1-0-3**

Course Objectives:

- To study the physics of wind power and energy
- To understand the principle of operation of wind generators
- To know the solar power resources and analyses of solar photo-voltaic cells
- To discuss the solar thermal power generation and identify the network integration issues

Course Outcomes:

At the end of this course, students will demonstrate the ability to

- Understand the energy scenario and the consequent growths of the power generate renewable energy sources.
- Understand the basic physics of wind and solar power generation.
- Understand the power electronic interfaces for wind and solar generation.
- Understand the issues related to the grid-integration of solar and wind energy systems

UNIT – I: Physics of Wind Power (8Hours)

History of wind power, Indian and Global statistics, Wind physics, Betz limit ratio, stall and pitch control, Wind speed statistics-probability distributions, and Wind power-cumulative distribution functions.

UNIT – II: Wind Generator Topologies (8Hours)

Review of modern wind turbine technologies, Fixed and Variable speed wind turbine, Induction Generators, Doubly-Fed Induction Generators and their characteristics, Permanent Magnet Synchronous Generators, Power electronics converters, Generator configurations, Converter Control.

UNIT – III: The Solar Resource (12Hours)

Introduction, solar radiation spectra, solar geometry, Earth Sun angles, observer Sun angles, solar day length, Estimation of solar energy availability.

Solar Photovoltaic

Technologies-Amorphous, mono-crystalline, polycrystalline; V-I characteristics of a PV cell, PV module, array, Power Electronic Converters for Solar Systems, Maximum Power point Tracking (MPPT) algorithms, Converter Control.

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UNIT – IV: Network Integration Issues (10Hours)

Overview of grid code technical requirements. Fault ride-through for wind farms - real and reactive power regulation, voltage and frequency operating limits, solar PV and wind farm behavior during grid disturbances. Power quality issues. Power system interconnection experiences in the world. Hybrid and isolated operations of solar PV and wind systems.

UNIT – V: Solar Thermal Power Generation (8Hours)

Technologies, Parabolic trough, central receivers, parabolic dish, Fresnel, solar pond, elementary analysis.

TEXT BOOKS:

1. T. Ackermann, "Wind Power in Power Systems", John Wiley and Sons Ltd., 2005.
2. G. M. Masters, "Renewable and Efficient Electric Power Systems", John Wiley and Sons, 2004.

REFERENCE BOOKS:

1. S. P. Sukhatme, "Solar Energy: Principles of Thermal Collection and Storage", McGraw Hill, 1984.
2. H. Siegfried and R. Waddington, "Grid integration of wind energy conversion systems" John Wiley and Sons Ltd., 2006.
3. G. N. Tiwari and M. K. Ghosal, "Renewable Energy Applications", Narosa Publications, 2004.
4. J. A. Duffie and W. A. Beckman, "Solar Engineering of Thermal Processes", John Wiley & Sons,1991.

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EE3222 DIGITAL CONTROL SYSTEMS

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- To understand the fundamentals of digital control systems, z-transforms
- To understand state space representation of the control systems, concepts of controllability and observability
- To study the estimation of stability in different domains
- To understand the design of discrete time control systems, compensators, state feedback controllers, state observers through various transformations

Course Outcomes:

At the end of this course, students will demonstrate the ability to

- Obtain discrete representation of LTI systems.
- Analyze stability of open loop and closed loop discrete-time systems.
- Design and analysis of digital controllers.
- Design state feedback and output feedback controllers.

UNIT- I: Discrete Representation of Continuous Systems (10Hours)

Basics of Digital Control Systems, Discrete representation of continuous systems. Sample and hold circuit, Mathematical Modelling of sample and hold circuit. Effects of Sampling and Quantization, Choice of sampling frequency, ZOH equivalent.

UNIT- II: Discrete System Analysis (12Hours)

Z-Transform and Inverse Z Transform for analyzing discrete time systems. Pulse Transfer function, Pulse transfer function of closed loop systems, Mapping from s-plane to z- plane, Solution of Discrete time system, Time response of discrete time system.

Stability of Discrete Time System: Stability analysis by Jury test. Stability analysis using bilinear transformation. Design of digital control system with dead beat response. Practical issues with dead beat response design.

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UNIT- III: State Space Approach for Discrete Time Systems (8Hours)

State space models of discrete systems, State space analysis. Lyapunov Stability. Controllability, reach-ability, Controllability and observability analysis. Effect of pole zero cancellation on the controllability & observability.

UNIT- IV: Design of Digital Control System (8Hours)

Design of Discrete PID Controller, Design of discrete state feedback controller. Design of set point tracker. Design of Discrete Observer for LTI System. Design of Discrete compensator.

UNIT- V: Discrete Output Feedback Control (8Hours)

Design of discrete output feedback control. Fast output sampling (FOS) and periodic output feedback controller design for discrete time systems.

TEXT BOOKS:

1. K. Ogata, "Digital Control Engineering", Prentice Hall, Englewood Cliffs, 1995.
2. M. Gopal, "Digital Control Engineering", Wiley Eastern, 1988.

REFERENCE BOOKS:

1. G. F. Franklin, J. D. Powell and M. L. Workman, "Digital Control of Dynamic Systems", Addison-
Wesley, 1998.
2. B.C. Kuo, "Digital Control System", Holt, Rinehart and Winston, 1980.

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EE3223 **ELECTRICAL DISTRIBUTION SYSTEMS**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-1-0-3**

Course Objectives:

- To distinguish between transmission and distribution systems
- To understand design considerations of feeders
- To compute voltage drop and power loss in feeders and understand protection of distribution systems
- To examine the power factor improvement and voltage control

Course Outcomes: After completion of this course, the student able to

- Distinguish between transmission, and distribution line and design the feeders
- compute power loss and voltage drop of the feeders
- Design protection of distribution systems
- Understand the importance of voltage control and power factor improvement

UNIT – I (12Hours)General Concepts:

Introduction to distribution system, Distribution system planning, Factors effecting the Distribution system planning, Load modelling and characteristics. Coincidence factor – contribution factor - Loss factor - Relationship between the load factor and loss factor. Load growth, Classification of loads (Residential, commercial, Agricultural and Industrial) and their characteristics.

Distribution Feeders: Design Considerations of Distribution Feeders: Radial, loop and network types of primary feeders, Introduction to low voltage distribution systems (LVDS) and High voltage distribution systems (HVDS), voltage levels, Factors effecting the feeder voltage level, feeder loading, Application of general circuit constants (A, B, C, D) to radial feeders, basic design practice of the secondary distribution system, secondary banking, secondary network types, secondary mains.

UNIT – II (10Hours)Substations:

Location of Substations: Rating of distribution substation, service area with 'n' primary feeders. Benefits derived through optimal location of substations. Optimal location of Substations (Perpendicular bisector rule and X, Y co-ordinate method).

System Analysis: Voltage drop and power-loss calculations: Derivation for voltage drop and power loss in lines, manual methods of solution for radial networks, three phase balanced primary lines, analysis of non-three phase systems, method to analyse the distribution feeder cost.

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UNIT – III (8Hours)

Protection: Objectives of distribution system protection, types of common faults and procedure for fault calculations, over current Protective Devices: Principle of operation of Fuses, Auto-Circuit Recloser - and Auto-line sectionalizes, and circuit breakers.

Coordination: Coordination of Protective Devices: Objectives of protection co-ordination, general coordination procedure, Types of protection coordination: Fuse to Fuse, Auto-Recloser to Fuse, Circuit breaker to Fuse, Circuit breaker to Auto-Recloser.

UNIT – IV (8Hours)

Compensation for Power Factor Improvement: Capacitive compensation for power-factor control - Different types of power capacitors, shunt and series capacitors, effect of shunt capacitors (Fixed and switched), effect of series capacitors, difference between shunt and series capacitors, Calculation of Power factor correction, capacitor allocation – Economic justification of capacitors - Procedure to determine the best capacitor location.

UNIT – V (8Hours)

Voltage Control: Voltage Control: Importance of voltage control, methods of voltage control, Equipment for voltage control, effect of shunt capacitors, effect of series capacitors, effect of AVB/AVR on voltage control, line drop compensation, voltage fluctuations.

TEXT BOOKS:

1. TuranGonen, Electric Power Distribution System Engineering, CRC Press, 3rd Edition 2014.

2. V. Kamaraju, Electrical Power Distribution Systems, Tata Mc Graw Hill Publishing Company,
2nd edition, 2010.

REFERENCE BOOKS:

1. G. Ram Murthy, Electrical Power Distribution hand book, 2nd edition, University press 2004.

2. A.S. Pabla, Electric Power Distribution, Tata McGraw Hill Publishing company, 6th edition, 2013

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EE3224 **SWITCH MODE POWER SUPPLIES**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives:

- This course introduces the basic concepts of switched mode power supplies, working of SMPS and different types of converters.
- The course discusses multiple output Fly back SMPS, uses of semiconductors in switched mode topology and different types of switched mode variable power supplies.

Course Outcomes: At the end of the course the student will be able to:

- Know the operation of SMPS, importance of semiconductors in SMPS and different types of SMPS
- Analyze different types of converters, the process of Rectification and different types of switched mode variable power supplies
- Switching devices - ideal and real characteristics, control, drive and protection.
- Reactive circuit elements - their selection and design.
- Switching power converters - circuit topology, operation, steady-state model, dynamic model.
- Analysis, modeling and performance functions of switching power converters.
- Review of linear control theory.
- Closed-loop control of switching power converters.
- Sample designs and construction projects.

UNIT I: Reactive circuit elements - their selection and design.: Introduction to DC-DC

converter, Diode Controlled Switches, Reactive components: Inductor, Transformer, Capacitor, Issues related to switches, Energy storage – Capacitor, Energy storage – inductor

UNIT II: Switched Mode Power Conversion (10 Hours)

Equivalent circuit model of the non-isolated DC-DC converters. Isolated converters (forward, Fly back).

Multiple Output Fly back Switch Mode Power Supplies (6 Hours)

Introduction, operating Modes, operating principles, Direct off line Flyback Switch Mode Power Supplies, Fly back converter, snubber network, Problems.

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UNIT III: Using Power Semiconductors in Switched Mode Topologies (8 Hours)
Half-Bridge and Full Bridge
Converters, Push-Pull Converter and SMPS with multiple outputs. Choice of switching frequency, The Power Supply Designer's Guide to High Voltage Transistors, Base Circuit Design for High Voltage Bipolar Transistors in Power Converters, Isolated Power Semiconductors for High Frequency Power Supply Applications

UNIT IV: Rectification (7 Hours)

Explanation, Advantages and disadvantages, SMPS and linear power supply comparison, Theory of operation , Input rectifier stage, Inverter stage, Voltage converter and output rectifier, Regulation, An Introduction to Synchronous Rectifier Circuits using Power MOS Transistors

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UNIT V: Concept of Zero Voltage switching (ZVS) and Zero current switching (ZCS) (6 Hours)

Resonant Power Supplies;
An Introduction to Resonant Power Supplies, Resonant Power Supply Converters - The Solution for Mains Pollution Problems.

Text Books:

1. "Switch Mode Power Supplies" by Keith H. Billings Taylor Morey- Tata McGraw-Hill Publishing Company, 3rd edition.
2. "Switch Mode Power Supplies", Robert W. Erickson.

Reference Books:

1. Switching Power Supplies A-Z, Second Edition- Sanjaya Maniktala.
2. Steven M. Sandler, Switch Mode Power Supplies, Tata McGraw Hill.

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EE3231 **HVDC TRANSMISSION**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

CourseObjectives:

- To compare EHV AC and HVDC systems
- To analyze Graetz circuit and also explain 6 and 12 pulse converters
- To control HVDC systems with various methods and to perform power flow analysis in AC/DCsystems
- To describe various protection methods for HVDC systems and Harmonics

CourseOutcomes:

- Compare EHV AC and HVDC system and to describe various types of DC links
- Analyze Graetz circuit for rectifier and inverter mode of operation
- Describe various methods for the control of HVDC systems and to perform power flow analysisin AC/DC systems
- Describe various protection methods for HVDC systems and classify Harmonics and designdifferent types of filters

UNITI: Basic Concepts & HVDC Converters (10Hours)

Necessity of HVDC systems, Economics and Terminal equipment of HVDC transmission systems, Types of HVDC Links, Apparatus required for HVDC Systems, Comparison of AC and DC Transmission, Application of DC Transmission System, Planning and Modern trends in D.C.Transmission.

Analysis of HVDC Converters: Choice of Converter Configuration, Analysis of Graetz circuit, Characteristics of 6 Pulse and 12 Pulse converters, Cases of two 3 phase converters in Y/Y mode & their performance.

UNITII: II Converter and Reactive Power Control (10Hours)

Principle of DC Link Control, Converters Control Characteristics, Firing angle control, Current and extinction angle control, Effect of source inductance on the system, Starting and stopping of DC link, Power Control.

Reactive Power Control in HVDC: Introduction, Reactive Power Requirements in steady state, sources of reactive power- Static VAR Compensators, Reactive power control during transients.

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UNIT III: Power Flow Analysis in AC/DC Systems **(8Hours)**

Modeling of DC Links, DC Network, DC Converter, Controller Equations, Solution of DC load flow, P.U. System for DC quantities, solution of AC-DC Power flow-Simultaneous method & Sequential method.

UNIT IV: Converter Faults and Protection**(8Hours)**

Converter faults, protection against over current and over voltage in converter station, surge arresters, smoothing reactors, DC breakers, Audible noise, space charge field, corona effects on DC lines, Radio interference

UNIT V: Harmonics & Filters(8Hours)

Generation of Harmonics, Characteristics of harmonics, calculation of AC Harmonics, Non-Characteristics harmonics, adverse effects of harmonics, Calculation of voltage and Current harmonics,Effect of Pulse number on harmonics

Filters: Types of AC filters, Design of Single tuned filters –Design of High pass filters.

TextBooks:

1. "K.R.Padiyar",HVDC Power Transmission Systems: Technology and system Interactions,New Age International (P) Limited, and Publishers, 1990.
2. "S K Kamakshaiah, V Kamaraju", HVDC Transmission, TMH Publishers, 2011
3. E. Uhlmann", Power Transmission by Direct Current, B. S. Publications, 2009

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EE3232 **VLSI DESIGN**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-1-0-3**

Course Objectives:

- Give exposure to different steps involved in the fabrication of ICs.
- Explain electrical properties of MOS and BiCMOS devices to analyze the behavior of inverters designed with various loads.
- Give exposure to the design rules to be followed to draw the layout of any logic circuit.
- Provide design concepts to design building blocks of data path of any system using gates.

Course Outcomes: After completion of this course, students will be able to

- Acquire qualitative knowledge about the fabrication process of integrated circuits using MOS transistors.
- Draw the layout of any logic circuit which helps to understand and estimate parasitic effect of any logic circuit
- Design building blocks of data path systems, memories and simple logic circuits using PLA, PAL, FPGA and CPLD.
- Understand different types of faults that can occur in a system and learn the concept of testing and adding extra hardware to improve testability of system.

UNIT-I: BASIC ELECTRICAL PROPERTIES (10 Hours)

Introduction to IC Technology – MOS, PMOS, NMOS, CMOS & BiCMOS.

Basic Electrical Properties of MOS and BiCMOS Circuits: Ids-Vds relationships, MOS transistor threshold Voltage, gm, gds, Figure of merit; Pass transistor, NMOS Inverter, Various pull ups, CMOS Inverter analysis and design, Bi-CMOS Inverters.

UNIT II: VLSI CIRCUIT DESIGN PROCESSES (8 Hours)

VLSI Design Flow, MOS Layers, Stick Diagrams, Design Rules and Layout, Transistors Layout Diagrams for NMOS and CMOS Inverters and Gates, Scaling of MOS circuits.

UNIT III: GATE LEVEL DESIGN (8 Hours)

Logic Gates and Other complex gates, Switch logic, Alternate gate circuits, Time delays, Driving large capacitive loads, Wiring capacitance, Fan – in, Fan – out.

UNIT IV:DATA PATH SUBSYSTEMS AND ARRAY SUBSYSTEMS (8 Hours)

Subsystem Design, Shifters, Adders, ALUs, Multipliers, Parity generators, Comparators, Zero/One Detectors, Counters.

SRAM, DRAM, ROM, Serial Access Memories.

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UNIT V: PROGRAMMABLE LOGIC DEVICES AND CMOS TESTING (8 Hours)

Design Approach – PLA, PAL, Standard Cells FPGAs, CPLDs.
CMOS Testing, Test Principles, Design Strategies for test, Chip level Test Techniques.

Text Books:

1. Essentials of VLSI circuits and systems – Kamran Eshraghian, Eshraghian Dougles and A. Pucknell, PHI, 2005 Edition
2. CMOS VLSI Design – A Circuits and Systems Perspective, Neil H. E Weste, David Harris, Ayan Banerjee, 3rd Ed, Pearson, 2009.

Reference Books:

1. Introduction to VLSI Systems: A Logic, Circuit and System Perspective – Ming-BO Lin, CRC Press, 2011
2. CMOS logic circuit Design - John. P. Uyemura, Springer, 2007.

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EE3233 **DIGITAL SIGNAL PROCESSING**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives: This course will develop student's knowledge in/on

- continuous-time (CT) and discrete-time (DT) signals
- discrete fourier transform (DFT), computational complexity of DFT and efficient implementation of DFT using fast fourier transform (FFT)
- specifying characteristics of frequency selective filters, design of linear-phase FIR filters
- classical analog butterworth & chebyshev filters, converting analog filter into equivalent digital filter to design digital IIR filters

Course Outcomes: At the end of the course the student will be able to:

- Represent signals mathematically in continuous and discrete-time, and in the frequency domain.
- Analyse discrete-time systems using z-transform.
- Understand the Discrete-Fourier Transform (DFT) and the FFT algorithms. Design digital filters for various applications.
- Apply digital signal processing for the analysis of real-life signals.

UNIT I: Discrete-time signals and systems (6 hours)

Discrete time signals and systems: Sequences; representation of signals on orthogonal basis; Representation of discrete systems using difference equations, Sampling and reconstruction of signals - aliasing; Sampling theorem and Nyquist rate.

UNIT II: Z-transform (6 hours)

Z-Transform, Region of Convergence, Analysis of Linear Shift Invariant systems using z-transform, Properties of z-transform for causal signals, Interpretation of stability in z-domain, Inverse z-transforms.

UNIT III: Discrete Fourier Transform (10 hours)

Frequency Domain Analysis, Discrete Fourier Transform (DFT), Properties of DFT, Convolutions of signals, Fast Fourier Transform Algorithm, Parseval's Identity, Implementation of Discrete Time Systems.

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UNIT IV: Design of Digital filters (12 hours)

Design of FIR Digital filters: Window method, Park-McClellan's method. Design of IIR Digital Filters: Butterworth, Chebyshev and Elliptic Approximations; Low-pass, Band-pass, Band- stop and High-pass filters. Effect of finite register length in FIR filter design. Parametric and non-parametric spectral estimation. Introduction to multi-rate signal processing.

UNIT V: Applications of Digital Signal Processing (6 hours)

Correlation Functions and Power Spectra, Stationary Processes, Optimal filtering using ARMA Model, Linear Mean-Square Estimation, Wiener Filter.

Text/Reference Books:

1. S. K. Mitra, " Digital Signal Processing: A computer based approach", McGraw Hill, 2011.
2. A.V. Oppenheim and R. W. Schafer, "Discrete Time Signal Processing" , Prentice Hall, 1989.
3. J. G. Proakis and D.G. Manolakis, " Digital Signal Processing: Principles, Algorithms And Applications", Prentice Hall, 1997.
4. L. R. Rabiner and B. Gold, "Theory and Application of Digital Signal Processing" , Prentice Hall, 1992.
5. J. R. Johnson, " Introduction to Digital Signal Processing" , Prentice Hall, 1992.
6. D. J. DeFatta, J. G. Lucas andW. S. Hodgkiss, " Digital Signal Processing" , John Wiley & Sons, 1988.

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EE3234 DESIGN OF PHOTOVOLTAIC SYSTEMS

Externals: 60Marks **L-T-P-C**
Internals: 40Marks **3-0-0-3**

Course Objectives:
The objectives of this course are:

1. Teach about the photovoltaic system design.
2. Discuss PV cell electrical characteristics and interconnections.
3. Enable students to know the concepts on MPPT and related circuits.
4. Enable students to learn about PV applications & grid interface.

UNIT-1: Introduction to Photovoltaic systems

PV cell characteristics and equivalent circuit, Model of PV cell, Short Circuit, Open Circuit and peak power parameter, Cell efficiency, Fill factor, Datasheet study, Effect of temperature, Temperature effect calculation example

UNIT-2: Series and Parallel Connection of Solar Cells

Identical cells in series, Load line, Non-identical cells in series, Protecting cells in series, Interconnecting modules in series, Identical cells in parallel, Non-identical cells in parallel, Protecting cells in parallel, Interconnecting modules in parallel, Insolation and Irradiance, Insolation variation with time of day, Atmospheric Partial shading effects

UNIT-3: MPPT Algorithms and PV Battery Interface

MPPT concept, Input impedance of DC-DC converters -Boost converter, Buck converter, Buck-Boost converter, Impedance control methods, Reference cell, Sampling method, Power slope methods, Hill climbing method, Practical points - Housekeeping power supply, Gate driver, Direct PV-battery connection, Charge controller, Battery charger - Understanding current control, slope compensation, Charge controller, battery charger, Batteries in series - charge equalization, Batteries in Parallel

UNIT-4: Peltier Cooling and Water Pumping Applications

Peltier device - principle, Peltier element - datasheet, Peltier cooling, Thermal aspects - Conduction, Convection , A peltier refrigeration example, Radiation and mass transport, Demo of Peltier cooling, Water pumping principle, Hydraulic energy and power, Total dynamic head, Centrifugal pump, Reciprocating pump, PV power, Pumped hydro application

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UNIT-5: PV Grid Interface

Grid connection principle, PV to grid topologies Part-I, PV to grid topologies Part-II, PV to grid topologies Part-III, 3 phase d-q controlled grid connection, 3 phase d-q controlled grid connection AC-DC transformations, 3 phase d-q controlled grid connection DC-AC transformations, 3 phase d-q controlled grid connection complete 3 phase grid connection, Single phase d-q controlled grid connection

Course Outcomes:

By the end of the course the students will be able to

1. Understood the photovoltaic system design.
2. Analyze and Design the PV cell electrical characteristics and interconnections.
3. Understood and ready to do the hard ware implementation of MPPT and related circuits.
4. Understood about the photovoltaic system applications & grid interface.

Books and references:

1. Chenming, H. and White, R.M., Solar Cells from B to Advanced Systems, McGraw Hill Book Co, 1983

2. Ruschenbach, HS, Solar Cell Array Design Hand Varmostrand, Reinhold, NY, 1980

3. Proceedings of IEEE Photovoltaics Specialists Conferences, Solar Energy Journal.

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EE4141 **SMART GRID**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- This course introduces the basic concepts of a Smart Grid, different types of constraints
- This course discusses different types of communication technologies for Smart Grid
- It introduces different types of Security systems for the Smart grid
- This course discusses the smart metering and distribution system modeling

Course Outcomes:

At the end of the course the students will be able to

- Understand the background for Smart Grid and have knowledge about important Terminologies
- Know about challenges and possibilities related to Smart Meters
- Analyze and perform basic design of Smart Grid electric power systems
- The course discusses the interaction between the power grid and Flexible resources and Smart Meters

UNITI: The Smart Grid (8Hours)

Introduction, Ageing Assets and Lack of Circuit Capacity, Thermal Constraints, Operational Constraints, Security of Supply, National Initiatives, Early Smart Grid Initiatives, Active Distribution Networks, Virtual Power Plant, Other Initiatives and Demonstrations, Overview of The Technologies Required for The Smart Grid.

UNITII: Communication Technologies (10Hours)

Data Communications: Introduction, Dedicated and Shared Communication Channels, Switching Techniques, Circuit Switching, Message Switching, Packet Switching, Communication Channels, Wired Communication, Optical Fibre, Radio Communication, Cellular Mobile Communication, Layered Architecture and Protocols, The ISO/OSI Model, TCP/IP Communication Technologies: IEEE802Series, Mobile Communications, Multi Protocol Label Switching, Power line Communication, Standards for Information Exchange, Standards For Smart Metering, Mod bus, DNP3,IEC61850.

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UNIT III: Information Security For The Smart Grid (8Hours)

Introduction, Encryption and Decryption, Symmetric Key Encryption, Public Key Encryption, Authentication, Authentication Based on Shared Secret Key, Authentication Based on Key Distribution Center, Digital Signatures, Secret Key Signature, Public Key Signature, Message Digest, Cyber Security Standards, IEEE1686: IEEE Standard for Substation Intelligent Electronic Devices(IEDs) Cyber Security Capabilities, IEC 62351: Power Systems Management And Association Information Exchange–Data and Communication Security.

UNIT IV: Smart Metering And Demand Side Integration (8Hours)

Introduction, smart metering–evolution of electricity metering, key components of smart metering, smart meters: an overview of the hardware used– signal acquisition, signal conditioning, analogue to digital conversion, computation, input/output, and communication.

Communication infrastructure and protocols for smart metering- Home are a network, Neighborhood Area Network, Data Concentrator, meter data management system, Protocols for communication. Demand Side Integration- Services Provided by DSI, Implementation of DSI, Hardware Support, Flexibility Delivered by Prosumers from the Demand Side, System Support from DSI.

UNIT V:Transmission And Distribution Management Systems (8Hours)

Data Sources, Energy Management System, Wide Area Applications, Visualization Techniques, Data Sources and Associated External Systems, SCADA, Customer Information System, Modeling and Analysis Tools, Distribution System Modeling, Topology Analysis, Load Forecasting, Power Flow Analysis, Fault Calculations, State Estimation, Applications, System Monitoring, Operation, Management, Outage Management System, Energy Storage Technologies, Batteries, Flow Battery, Fuel Cell and Hydrogen Electrolyses, Flywheels, Superconducting Magnetic Energy Storage Systems, Super capacitors.

Text Books:

1. Smart Grid, Janaka Ekanayake, Liyanage, Wu, AkihikoYokoyama, Jenkins, Wiley Publications, 2012.
2. Smart Grid: Fundamentals of Design and Analysis. James Momoh,Wiley, IEEE Press., 2012.

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EE4142

ARTIFICIAL NEURAL NETWORKS AND FUZZY SYSTEMS

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- To introduce the basics of Neural Networks and its architectures.
- To introduce the Fuzzy sets and Fuzzy Logic system components.
- To deal with the applications of Neural Networks.
- To deal with the applications of Fuzzy systems.

Course Outcomes: At the end of the course the students will be able to

- To understand artificial neural network models and their training algorithms.
- To understand the concept of fuzzy logic system components, fuzzification and defuzzification.
- Design the layout of Gas Insulated substations.
- To know the concepts of insulation coordination.

UNIT I: Introduction To Neural Networks and Essentials of Artificial Neural Networks (9Hours)

Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin- Huxley Neuron Model, Integrate- and- Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCulloch- Pitts Model, Historical Developments, Potential Applications of ANN.

Essentials of Artificial Neural Networks

Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN–Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application.

UNIT II: Feed forward and Multi layer Feed forward neural networks (9Hours)

Single Layer Feed Forward Neural Networks: Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. **Multi layer Feed forward**

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Neural Networks

Credit Assignment Problem, Generalized Delta Rule, Derivation of Back propagation (BP) Training, Summary of Back propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements

UNIT III: Associative Memories (8Hours)

Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).

Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms

Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hop field Network.

UNIT IV: Classical And Fuzzy Sets (7Hours)

Introduction to classical sets-properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.

UNIT V: Fuzzy Logic System (7Hours)

Fuzzification, Membership value assignment, development to frulebase and decision making system, Defuzzification to crisp sets, Defuzzification methods.

Text books:

1. Rajasekharan and Pai, Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications–PHI Publication.
2. Satish Kumar , Neural Networks, TMH, 2004.

Reference books:

1. James A Freeman and Davis Skapura, Neural Networks, Pearson Education, 2002.
2. Simon Hakins, Neural Networks, Pearson Education.
3. C. Elia smith and Ch. Anderson, Neural Engineering, PHI.

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EE4143 **FLEXIBLE AC TRANSMISSION**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **4-0-0-3**

Course Objectives:

- To understand the concept of flexible AC transmission and the associated problems.
- To review the static devices for series and shunt control.
- To study the operation of controllers for enhancing the transmission capability.

Course Outcomes: Upon the completion of this course, the student will be able:

- To apply knowledge of FACTS Controllers.
- To design a Compensators within realistic constraints.
- To identify, formulate, and solve real network problems with FACTS controllers

UNIT I: **INTRODUCTION**

The concept of flexible AC transmission - reactive power control in electrical power transmission lines -uncompensated transmission line – series and shunt compensation. Overview of FACTS devices - Static Var Compensator (SVC) – Thyristor Switched Series capacitor (TCSC) – Unified Power Flow controller (UPFC) - Integrated Power Flow Controller (IPFC).

UNIT II: STATIC VAR COMPENSATOR (SVC) AND APPLICATIONS

Voltage control by SVC – advantages of slope in dynamic characteristics – influence of SVC on system voltage. Applications - enhancement of transient stability – steady state power transfer – enhancement of power system damping – prevention of voltage instability.

UNIT III: THYRISTOR CONTROLLED SERIES CAPACITOR (TCSC) AND APPLICATIONS

Operation of the TCSC - different modes of operation – modeling of TCSC – variable reactance model – modeling for stability studies. Applications - improvement of the system stability limit – enhancement of system damping – voltage collapse prevention.

UNIT IV: EMERGING FACTS CONTROLLERS

Static Synchronous Compensator (STATCOM) – operating principle – V-I characteristics Unified Power Flow Controller (UPFC) – Principle of operation - modes of operation – applications – modeling of UPFC for power flow studies.

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UNIT V: CO-ORDINATION OF FACTS CONTROLLERS

FACTs Controller interactions – SVC–SVC interaction - co-ordination of multiple controllers using linear control techniques – Quantitative treatment of control coordination. Page 86 Electrical and Electronics Engineering.

Text Books

1. Hingorani ,L.Gyugyi, 'Concepts and Technology of Flexible AC Transmission System', IEEE Press New York, 2000

2. Mohan Mathur, R., Rajiv. K. Varma, "Thyristor – Based Facts Controllers for Electrical Transmission Systems", IEEE press and John Wiley & Sons, Inc.

REFERENCES:

1. A.T.John, "Flexible AC Transmission System", Institution of Electrical and Electronic Engineers (IEEE), 1999.
2. Narain G.Hingorani, Laszio. Gyugyl, "Understanding FACTS Concepts and Technology of Flexible AC Transmission System", Standard Publishers, Delhi 2001

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EE4144 **IoT in Micro grid** L-T-P-C

3-0-0-3

Course Objective:

• To provide the basic concepts of Micro grid.

To provide configuration of Micro grid

• To provide configuration of Micro grid operation and control.

To familiarize the students with energy storage devices, smart metering and loT application in Micro grid.

Course Outcomes: At the end of this course, students will demonstrate the ability to

Understand overview of micro grid and its operation & control.

Understand energy storage system and their application in micro grid.

Understand Architecture of lot and Communication network.

Understand Smart Meters and Smart sensors.

Unit I. An Overview of Micro grid

Concept of Micro grid, Typical structure and configuration of a Micro grid, Significance of Micro grid, Sources of micro grid, Types of Micro grids, AC, DC and hybrid Micro grids.

Unit II. Microg rid Operation and Control

Modes of Operation: Grid Connected Mode, Islanding Mode, Issues in Island Mode of operations, Control laws, Power relations and power control, Bi-directionality and its need in Micro grid, Control of DC-DC converters and inverter and challenges in a Micro grid, Micro grid Control Strategies: Centralized, Decentralized and Hierarchical control.

Unit III. Energy Storage for Micro grid

Role of energy storage systems and their applications inMicro grid, Overview of energy storage

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technologies: Thermal, Mechanical, Chemical, Electrochemical, Electrical, Battery Energy Storage Systems (BESS), Superconducting Magnetic Energy Storage (SMES), Compressed Air Energy Storage (CAES).

Unit IV. Introduction to IoT

Architecture of IoT, Communication network: Home Area Network (HAN), Neighborhood Area Network (NAN), Field Area Network (FAN), Wide Area Network (WAN), Wireless Sensor Networks (WSNs).

Unit V. IoT in Microgrid

Smart Meters, Automatic Meter Reading (AMR), Advanced Metering Infrastructure (AMI), Real Time Pricing, Smart Appliances. Smart sensors: home & building automation, plug in hybrid electric-vehicles (PHEV), algorithms for vehicle to grid and grid to vehicle management, smart charging stations.

Text Books:

2. Micro grids: Architectures and Control by Nikos Hatziargyriou, Wiley-IEEE Press, 2013.
3. Micro grid: Advanced Control Methods and Renewable Energy System Integration by Magdi SMahmoud, Butterworth-Heinemann, 2016
4. Microgrids and Active Distribution Networks by S. Chowdhury, P. Crossley, IET Press, 2010
5. Design of Smart Power Grid Renewable Energy Systems, AliKeyhani, John Wiley & Sons, 2011.
6. Smart Grid: Infrastructure, Technology and Solutions by Stuart Borlase, CRC Press 2012.
7. Smart Grid: Technology and Applications by Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Jianzhong Wu, Akihiko Yokoyama, Wiley

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EE4151 **POWER QUALITY**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives

- To know different terms of power quality.
- To Illustrate of voltage power quality issue – short and long interruption
- To construct study of characterization of voltage sag magnitude and three phase unbalanced voltage sag.
- To know the behavior of power electronics loads; induction motors, synchronous motor etc by the power quality issues

Course Outcomes: Upon the completion of the subject, the student will be able to

- Know the severity of power quality problems in distribution system;
- Understand the concept of voltage sag transformation from up-stream (higher voltages) to down-stream (lower voltage)
- compute the concept of improving the power quality to sensitive load by various mitigating custom power devices
- prepare mitigation of power quality issues by the VSI converters.

Unit-I: Introduction

Introduction of the Power Quality (PQ) problem, Terms used in PQ: Voltage, Sag, Swell, Surges, Harmonics, over voltages, spikes, Voltage fluctuations, Transients, Interruption, overview of power quality phenomenon, Remedies to improve power quality, power quality monitoring.

Unit-II: Long & Short Interruptions

Interruptions – Definition – Difference between failures, outage, Interruptions – causes of Long Interruptions – Origin of Interruptions – Limits for the Interruption frequency – Limits for the interruption duration – costs of Interruption – Overview of Reliability evaluation to power quality, comparison of observations and reliability evaluation. Short interruptions: definition, origin of short interruptions, basic principle, fuse saving, voltage magnitude events due to re-closing, voltage during the interruption, monitoring of short interruptions, difference between medium and low voltage systems. Multiple events, single phase tripping – voltage and current during fault period, voltage and current at post fault period, stochastic prediction of short interruptions

Unit III: Single & Three-Phase Voltage SAG Characterization

Voltage sag – definition, causes of voltage sag, voltage sag magnitude, and monitoring, theoretical calculation of voltage sag magnitude, voltage sag calculation in non-radial systems, meshed systems, and voltage sag duration. Three phase faults, phase angle jumps, magnitude and phase angle jumps for three phase unbalanced sags, load influence on voltage sags.

Unit-IV: Power Quality Considerations in Industrial Power Systems

Voltage sag – equipment behavior of Power electronic loads, induction motors, synchronous motors, computers, consumer electronics, adjustable speed AC drives and its operation. Mitigation of AC Drives, adjustable speed DC drives and its operation, mitigation methods of DC drives.

Unit-IV: Power Quality Considerations in Industrial Power Systems

Voltage sag – equipment behavior of Power electronic loads, induction motors, synchronous motors, computers, consumer electronics, adjustable speed AC drives and its operation. Mitigation of AC Drives, adjustable speed DC drives and its operation, mitigation methods of DC drives.

Unit-V: Mitigation of Interruptions & Voltage Sags

Overview of mitigation methods – from fault to trip, reducing the number of faults, reducing the fault clearing time changing the power system, installing mitigation equipment, improving equipment immunity, different events and mitigation methods. System equipment interface – voltage source converter, series voltage controller, shunt controller, combined shunt and series controller. Power Quality and EMC Standards: Introduction to standardization, IEC Electromagnetic compatibility standards, European voltage characteristics standards, PQ surveys.

TEXTBOOKS:

1. Math H J Bollen “Understanding Power Quality Problems”, IEEE Press.
2. R.C. Dugan, M.F. McGranaghan and H.W. Beaty, “Electric Power Systems Quality.” New York: McGraw-Hill. 1996

REFERENCES:

1. G.T. Heydt, ‘Electric Power Quality’, 2nd Edition. (West Lafayette, IN, Stars in a Circle Publications, 1994).
2. Power Quality VAR Compensation in Power Systems, R. SastryVedamMulukutla S. Sarma, CRC Press.
3. A Ghosh, G. Ledwich, Power Quality Enhancement Using Custom Power Devices. Kluwer Academic, 2002

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EE4152 **INDUSTRIAL AUTOMATION AND CONTROL**

Externals: 60Marks **L-T-P-C**

Internals: 40Marks **3-0-0-3**

Course Objectives:

The course should enable the students to :

- Learn the fundamental concepts about introduction to industrial automation and control and devices.
- Study the performance of each system in detail along with practical case studies.
- Develop various types of industrial automation and control and devices.
- Understand the process control of PLC automation.

Course Outcomes:

- Learn the control architecture of industrial automation system
- Understand the process control, PID control, controller tuning
- Understand the sequential function charts, the PLC hardware environment
- Develop the knowledge on CNC machines and actuators

UNIT-I: INTRODUCTION TO INDUSTRIAL AUTOMATION AND CONTROL (08 Hours)

Introduction to Industrial Automation and Control, control architecture of industrial automation system, measurement systems specifications, temperature measurement, pressure and force measurement, displacement and speed measurement, signal conditioning circuits, errors and calibration.

UNIT-II : PROCESS CONTROL (10 Hours)

Introduction to process control, PID control, controller tuning, implementation of PID controllers, special control structures, feed forward and ratio control special control structures: predictive control, control of systems with inverse response.

UNIT-III : PROGRAMMABLE LOGIC CONTROL SYSTEMS (09 Hours)

Introduction to sequence or logic control and programmable logic controllers, the software environment and programming of PLCs, formal modeling of sequence control specifications, Programming.

programming of PLCs: sequential function charts, the PLC hardware environment

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UNIT-IV: CNC MACHINES AND ACTUATORS (10 Hours)

CNC machines and actuators: Introduction to computer numerically controlled machines, control valves, hydraulic actuation systems, principle and components, directional control valves, switches and gauges, industrial hydraulic circuits

UNIT-V : ELECTRICAL MACHINE DRIVES(08 Hours)

Electrical machine drives: Energy savings with variable speed drives, step motors: principles, construction and drives, electrical actuators, DC motor drives, electrical actuators: induction motor drives, electrical actuators, BLDC motor drives.

Text Books:

1. Madhu Chanda Mitra, Samarjit Sen Gupta**, "**Programmable Logic Controllers and Industria lAutomation:AnIntroduction", PenramInternationalPublishing(India) Pvt. Ltd**.**,1st Edition, 2008.
2. K Krishna swamy, S Vijayachitra, "IndustrialInstrumentation", New Age Publications, 1st Edition,2010.
3. Rajesh Mehra, Vikrant Vij**, "**PLCs & SCADA: Theory and Practice", Laxmi publications, 2nd Edition,2016

Reference Books:

1. A.K.Gupta, SK Arora**, "**Industrial Automation andRobotics", Laxmi Publications, 2nd Edition, 2013.
2. Jon Stenerson,"IndustrialAutomationandProcessControl",PrenticeHall,1st Edition,2002.

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EE4153 **HIGH VOLTAGE ENGINEERING**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objectives:

- To develop knowledge on generation of high voltage DC, AC (power frequency and high frequency), impulse voltages and currents
- To know the measurement of high voltages DC, AC (power frequency and high frequency), impulse voltages and currents
- To understand thoroughly various high voltage testing techniques of power apparatus and Insulation coordination in power systems

Course Outcomes: At the end of the course the student will be able to:

- Understand breakdown phenomena in gases and to elucidate the concepts used for the generation of high voltages and currents.
- Elucidate the concepts used for the measurement of high voltages and currents and design Corresponding circuits.
- Understand high voltage testing techniques of Power apparatus and causes of over voltage in Power systems.
- Design the layout of Gas Insulated substations and to know the concepts of insulation coordination.

UNIT I: Introduction To High Voltage Technology And Applications (7 Hours)

Electric Field Stresses, Gas / Vacuum as Insulator, Liquid Dielectrics, Solids and Composites, Estimation and Control of Electric Stress, Numerical methods for electric field computation, Surge voltages, their distribution and control, Applications of insulating materials in transformers, rotating machines, circuit breakers, cable power capacitors and bushings.

UNIT II: Break Down in Solid Dielectrics Gaseous and Liquid Dielectrics(8 Hours)

Gases as insulating media, collision process, Ionization process, Townsend's criteria of breakdown in gases, Paschen's law - Liquid as insulator, pure and commercial liquids - breakdown in pure and commercial liquids.

Break Down In Solid Dielectrics

Intrinsic breakdown, electromechanical breakdown, thermal breakdown, breakdown of solid dielectrics in practice, Breakdown in composite dielectrics, solid dielectrics used in practice.

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UNIT III: Generation of Measurement High Voltages And Currents(9 Hours)

Generation of High Direct Current Voltages, Generation of High alternating voltages, Generation of Impulse Voltages, Generation of Impulse currents, Tripping and control of impulse generators.

Measurement Of High Voltages And Currents

Measurement of High Direct Current voltages, Measurement of High Voltages alternating and impulse, Measurement of High Currents-direct, alternating and Impulse, Oscilloscope for impulse voltage and current measurements.

UNIT IV :Non-Destructive Testing Of Material And Electrical Apparatus(7Hours)

Measurement of D.C Resistivity, Measurement of Dielectric Constant and loss factor, Partial discharge measurements.

High Voltage Testing Of Electrical Apparatus: Testing of Insulators and bushings, Testing of Isolators and circuit breakers, testing of cables, Testing of Transformers, Testing of Surge Arresters, and Radio Interference measurements**.**

UNIT V: Over Voltage Phenomenon And Insulation Co-Ordination (7 Hours)

Natural causes for over voltages – Lightning phenomenon, Overvoltage due to switching surges, system faults and other abnormal conditions, Principles of Insulation Coordination on High voltage and Extra High Voltage power systems.

Text books:

1. M.S.Naidu and V. Kamaraju , High Voltage Engineering by– TMH Publications, 3rd Edition
2. E.Kuffel, W.S.Zaengl, J.Kuffel , High Voltage Engineering: Fundamentals by Elsevier, 2nd Edition.

Reference books:

1. C.L.Wadhwa , High Voltage Engineering by, New Age Internationals (P) Limited, 1997.
2. RavindraArora, Wolfgang Mosch, High Voltage Insulation Engineering by, New Age International (P) Limited, 1995.
3. Mazen Abdel Salam, Hussein Anis, Ahdan El-Morshedy, Roshdy Radwan , Marcel Dekker High Voltage Engineering, Theory and Practice.

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EE4154 **HYBRID ELECTRICAL VEHICLES**

Externals:60Marks **L-T-P-C**

Internals:40Marks **3-0-0-3**

Course Objective:

- To present a comprehensive overview of Electric and Hybrid Electric Vehicles
- To deliver and discuss power electronics based drive control systems
- To discuss the battery management systems.
- To discuss grid integration issues of Electric and Hybrid vehicles.

Course Outcomes: At the end of the course the student will be able to:

- Understand the models to describe hybrid vehicles and their performance.
- Understand the different possible ways of energy storage.
- Understand the different strategies related to energy storage systems.
- Analyze and model the power management systems for electric and hybrid vehicles

UNIT I: Introduction to Conventional Vehicles and Hybrid Electric Vehicles : (10 hours)

Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, and mathematical models to describe vehicle performance.

Introduction to Hybrid Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies.

UNIT II: Hybrid Electric Drive-trains: (6 hours)

Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.

UNIT III: Electric Trains (10 hours)

Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive- train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis. Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and

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control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency.

UNIT IV: Energy Storage (9 hours)

Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Flywheel based energy storage and its analysis, Hybridization of different energy storage devices. Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE).

UNIT V: Energy Management Strategies (7 hours)

Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies.
Case Studies: Design of a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV).

TEXT BOOKS

1. C. Mi, M. A. Masrur and D. W. Gao, " Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives", John Wiley & Sons, 2011.
2. S. Onori, L. Serrao and G. Rizzoni, " Hybrid Electric Vehicles: Energy Management Strategies", Springer, 2015.

Reference Books:

1. M. Ehsani, Y. Gao, S. E. Gay and A. Emadi, " Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design", CRC Press, 2004.
2. T. Denton, "Electric and Hybrid Vehicles", Routledge, 2016.

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